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SCSE2243 APPLICATION DEVELOPMENT I

SECTION 01

SOFTWARE REQUIREMENT SPECIFICATION

UTM BORROW

A Campus-Wide Circular Resource Sharing Platform

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1. INTRODUCTION TO APPLICATION

1.1. Introduction

UTM Borrow is a mobile application designed specifically as a circular resource-sharing platform for the Universiti Teknologi Malaysia (UTM) community. It aims to facilitate a secure, centralized environment where students and staff can lend and borrow academic resources such as laboratory tools, engineering equipment, textbooks, and event clothing. Unlike unverified WhatsApp or Telegram groups, UTM Borrow introduces strict trust mechanics to ensure accountability. The app promotes campus sustainability by reducing post-semester disposal waste and supports students who face financial constraints when purchasing expensive, single-use tools.

1.2. Problem Statement

The Universiti Teknologi Malaysia (UTM) student community currently faces significant financial and logistical challenges when attempting to acquire specialized academic and extracurricular resources. Many university courses require students to purchase high-cost, single-use items—such as scientific laboratory equipment, engineering drawing tools, formal presentation attire, and specialized electronics—that are often utilized for only a single semester or specific project. This "buy-and-discard" culture imposes a disproportionate financial burden on students from lower-income backgrounds, exhausts limited student budgets, and generates immense post-semester environmental waste across residential colleges.

Presently, students attempt to mitigate these costs by relying on informal peer-to-peer sharing through unverified communication channels like WhatsApp and Telegram. However, these unstructured platforms completely lack an official identity verification system and a formalized trust infrastructure. Consequently, students are highly vulnerable to scams and theft by non-university affiliates. Furthermore, there is a profound absence of accountability mechanisms for item returns and condition tracking. Lenders frequently face disputes over damaged items due to the lack of pre-handover condition logs, and items are regularly returned late or entirely forgotten due to the absence of automated tracking and digital return deadlines. The inherent risks associated with these informal channels actively discourage resource owners from lending their valuable assets, effectively preventing the formation of a sustainable, campus-wide circular economy.

1.3. Technology and Tool Used

UTM Borrow is engineered as a responsive web application built on a modern, three-tier client–server stack. Every technology below reflects the system as actually built and deployed; selections prioritise security, real-time responsiveness, and a single maintainable codebase.

1. Frontend Development (Client-Side):

- **React (Create React App):** The user interface is a single-page application (SPA) built in React, delivering a fast, responsive experience across desktop and mobile browsers from one unified codebase — no separate native builds **required.**
- **Tailwind CSS:** A utility-first styling framework used to compose the entire interface, ensuring a consistent, professional, and fully responsive visual design.
- **Framer Motion:** Drives the interface animations and transitions (modals, status changes, page transitions), giving the platform a polished, app-like feel.
- **Axios:** The HTTP client used to communicate with the backend REST API over secure, token-authenticated requests.

2. Backend Development & Database (Server-Side):

- **FastAPI (Python):** A high-performance asynchronous web framework serving the complete REST API (all routes namespaced under /api). It enforces request validation, authentication, and the transaction state-machine logic on the server.
- **MongoDB (via the Motor async driver):** A document-oriented NoSQL database serving as the core data repository. It stores user profiles, the item catalog, and the full transaction lifecycle as JSON-style documents, each keyed by a unique string identifier.
- **JWT (HS256) Authentication + bcrypt:** Access is gated by JSON Web Tokens (7-day validity) issued only to verified UTM members; passwords are never stored in plain text — they are hashed with the bcrypt algorithm. A regular-expression gate restricts registration to @utm.my / @graduate.utm.my addresses.
- **HMAC-SHA256 Signed QR Tokens:** Physical handover and return are proven using cryptographically signed, time-limited (24-hour) QR tokens in the format UTMB.<payload>.<signature>, verified server-side on every scan.
- **Server-Sent Events (SSE):** A one-way server→client event stream pushes live updates (new notifications, catalog changes, transaction updates, chat messages, moderation events) to the browser over plain HTTP without polling, replacing the need for a third-party push service.

- **TOTP Multi-Factor Authentication (Admin):** Administrative moderation actions are protected by authenticator-app 6-digit codes that mint a short-lived elevated admin session.

3. Deployment Infrastructure:

- Render hosts the FastAPI backend as a Docker container (python:3.12-slim, served by uvicorn); Vercel hosts the React production build. The two services are wired together through environment variables and a CORS policy.

4. System Modeling, Design, and Project Management:

- **Visual Paradigm:** The mandatory modeling tool used to produce all Unified Modeling Language (UML) artifacts — Use Case Diagrams, Sequence Diagrams, and Class Diagrams — mapping the logical flow of all three subsystems.
- **Figma:** Used during requirement engineering to construct high-fidelity UI/UX wireframes and interactive prototypes ahead of frontend implementation.
- **Jira & GitHub:** Jira tracks weekly deliverables and sprint progress on an agile board; GitHub serves as the centralised version-control repository for source code and collaborative integration.
- **Generative AI (ChatGPT / Claude):** Used as a development-acceleration aid for boilerplate generation, documentation drafting, and debugging support throughout the project.

1.4. Functional Requirement

The platform's functional capabilities are distributed across nine modules within three subsystems, as summarised below. Detailed, testable requirements for each module are specified in their respective subsystem sections

2. APPLICATION DIAGRAM

2.1. Use Case Diagram

3. Identity & User Management Subsystem - Ahmat

3.1. Subsystem Overview

The Identity & User Management Subsystem is the foundational security and accountability layer of the UTM Borrow platform. It governs how students access the system, establishes their digital identity, and maintains a transparent track record of their borrowing and lending behaviors. By tightly integrating with official university credentials, this subsystem ensures that the sharing ecosystem remains closed, secure, and exclusive to active members of the Universiti Teknologi Malaysia community.

3.1.1. Purpose

The primary purpose of this subsystem is to authenticate users, manage their public profiles, and provide a centralized dashboard for tracking ongoing commitments. Unlike unverified social media chat groups, this subsystem introduces structured identity verification and a community-driven trust mechanism. It guarantees that every user is held accountable for their actions, providing lenders with the confidence they need to list their valuable academic resources.

3.1.2. Problem Statement

Currently, the campus resource-sharing environment relies heavily on informal, unverified chat groups such as WhatsApp or Telegram. Because these platforms lack an official identity verification system linked to UTM credentials, students are highly vulnerable to scams, false damage claims, and theft by non-affiliates. Furthermore, the profound absence of a trust profile and reputation system actively discourages owners from lending their valuable items, as there is no reliable way to verify a potential borrower's track record.

3.1.3. Subsystem Objectives

Implement UTM Email Verification: Restrict platform access exclusively to users with valid @graduate.utm.my or @utm.my email addresses to prevent external scammers.

Establish a Reputation System: Create a dynamic Profile and Trust Score rating system that automatically updates after every transaction to encourage honest behavior and highlight reliable community members.

Centralize User Activity: Build a User Activity Dashboard that provides a real-time summary of all active lending and borrowing commitments to prevent missed return deadlines and lost items.

3.1.4. SUBSYSTEM ARCHITECTURE OVERVIEW

3.1.4.1. Modules Structure

The Identity & User Management Subsystem is organized into three integrated modules, each responsible for a distinct phase of the transaction lifecycle:

Module	Purpose & Scope
Authentication Module	Manages user registration, login, and password recovery using official UTM student emails to guarantee only verified university members can access the platform.
Profile & Trust Score	Manages user profile details and implements a 5-star rating system (Trust Score) updated after every transaction. Enables lenders to file official non-compliance reports for UTM Administration review.
User Activity Dashboard	Serves as a central hub providing a quick summary of "Items I am Lending" and "Items I am Borrowing" so users never lose track of their active commitments.

3.1.4.2. Use Case Diagram

The following diagram illustrates the actors and their interactions across all three modules of the The Identity & User Management Subsystem

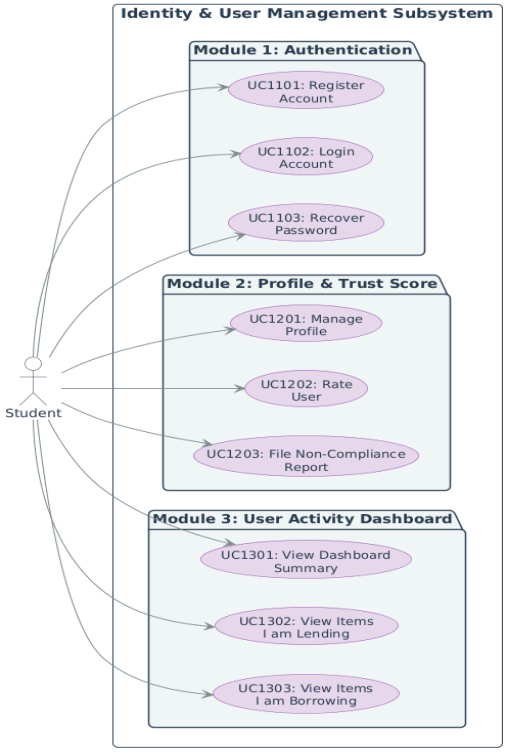


Figure 3.1.4: Use Case Diagram of the Identity & User Management Subsystem Architecture

3.2. Module 1: Authentication

3.2.1. Structure of Module 1:

The Authentication Module is the primary gatekeeper of the platform. It ensures that every user entering the platform is an active, verified member of the UTM community by strictly gating registration behind official university email domains.

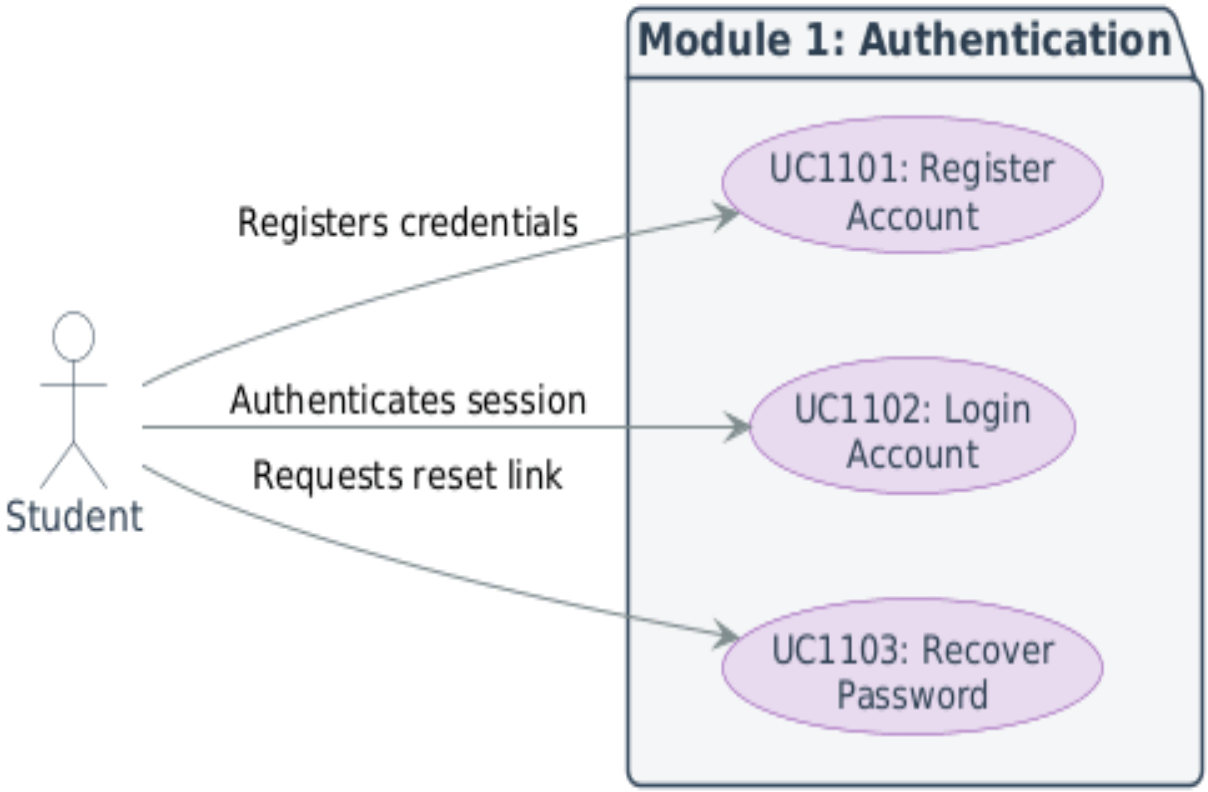


Figure 3.2: Package Diagram of Authentication Module

3.2.2. Specific Requirements

Registration must strictly enforce format validation for `@graduate.utm.my` or `@utm.my` email domains. Passwords must be cryptographically hashed and encrypted before being stored in the shared Firebase database. The system must restrict duplicate email registrations and provide a secure, token-based recovery system for forgotten passwords.

3.2.3. UC1101: Register Account

3.2.3.1. Use Case Specifications

Use Case: Register Account	ID: UC1101
Actors	Student Borrower, Student Lender, System
Module	Authentication

Preconditions

1. The student does not have an existing account in the UTM Borrow system.
2. The student possesses a valid, active UTM email address.

Flow of Events

1. The student navigates to the Registration screen upon launching the application.
2. The student enters their UTM email, matric number, and a secure password.
3. The student taps the "Sign Up" button.
4. The system validates the email input to ensure it ends with the official UTM domain.
5. The system queries the database to confirm the email is not already registered.
6. The system records the new user in the database and encrypts the password securely.
7. The system displays a success message: "Account created successfully. Welcome to UTM Borrow!"
8. The system automatically navigates the student back to the Login screen.

Alternate Flows

A1 — Email Already Registered:

1. At step 5, the system detects that the email is already present in the database.
2. The system blocks registration and displays an error alert: "This email is already registered. Please proceed to login."

A2 — Invalid Email Format:

1. At step 4, the system detects a non-UTM email address (e.g., @gmail.com).
2. The system displays a validation error: "Only official UTM student emails are allowed for registration."

Post-conditions

1. A new, verified user account is successfully created and permanently stored in the system database.

Table 3.2.3: Use Case Description of UC1101 — Register Account

3.2.3.2. Sequence Diagram

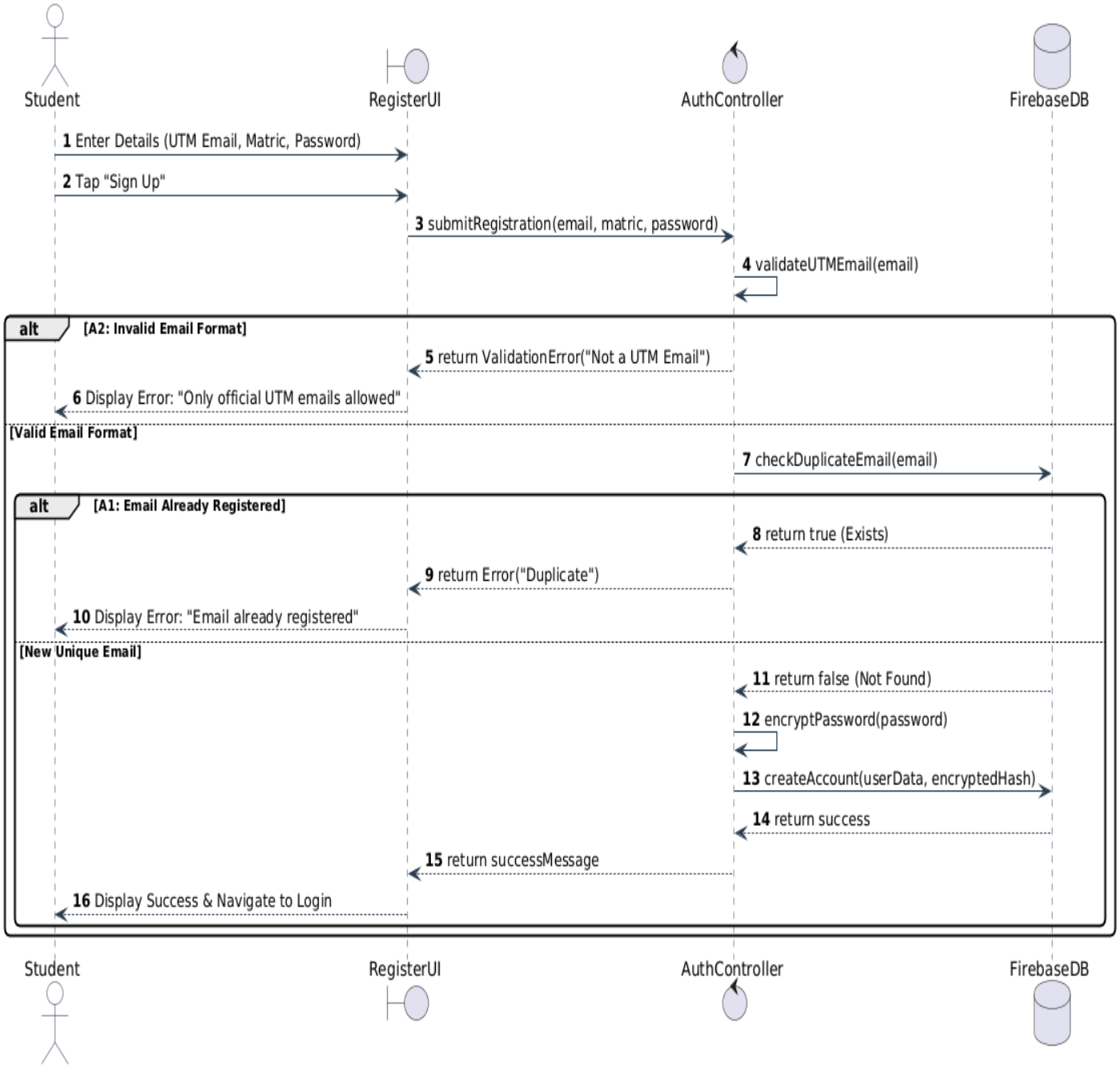


Figure 3.2.3: Sequence Diagram of UC1101 — Register Account

3.2.4. UC1102: Login Account

3.2.4.1. Use Case Specifications

Use Case: Login Account	ID: UC1102
Actors	Student Borrower, Student Lender, System
Module	Authentication
Preconditions	
1. The user has successfully completed the registration process and possesses an active account.	
Flow of Events	
1. The student opens the app and navigates to the Login screen. 2. The student inputs their registered UTM email and password. 3. The student taps the "Login" button. 4. The system securely transmits the credentials and verifies them against the encrypted database records. 5. The system confirms a successful match and generates a secure user session token. 6. The system grants access and navigates the user directly to the Resource Catalog.	
Alternate Flows	
A1 — Incorrect Credentials: 1. At step 4, the system detects a mismatch between the provided email and the stored password hash. 2. The system denies access and displays an error message: "Invalid email or password. Please try again."	
Post-conditions	
1. An active user session is securely established, allowing the user to interact with the platform's core features.	

Table 3.2.4: Use Case Description of UC1102 — Login Account

3.2.4.2. Sequence Diagram

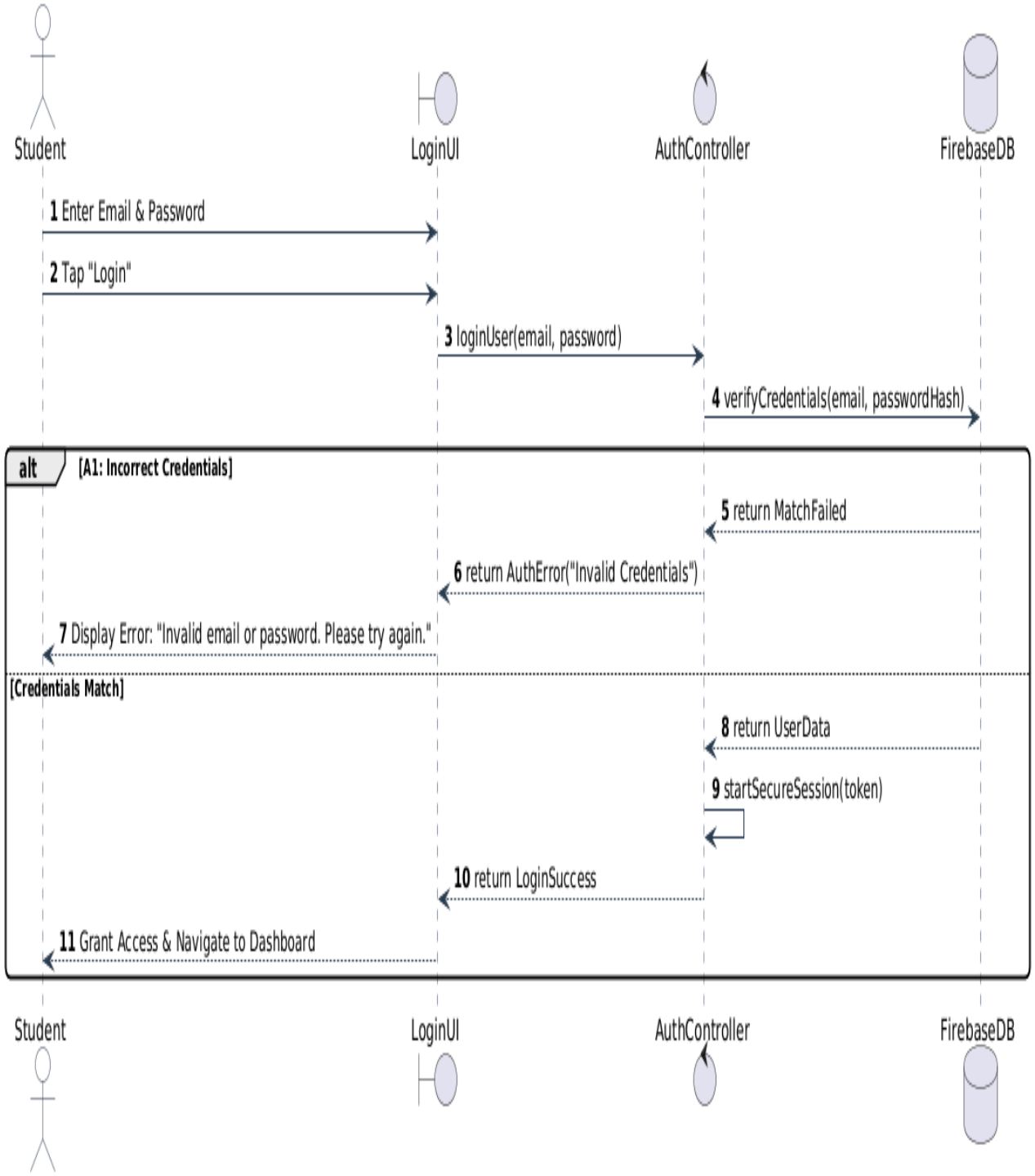


Figure 5.2.5: Sequence Diagram of UC1102 — Login Account

3.2.5. UC1103: Recover Password

3.2.5.1. Use Case Specifications

Use Case: Recover Password	ID: UC1103
Actors	Student Borrower, Student Lender, System
Module	Authentication

Preconditions

1. The user has forgotten their password but still retains access to their UTM email inbox.

Flow of Events

1. The student taps the "Forgot Password" link on the main Login screen.
2. The system displays a prompt requiring the student to enter their registered UTM email.
3. The student enters their email and taps "Send Recovery Link".
4. The system verifies that the email exists within the database.
5. The system automatically sends a secure, time-sensitive password recovery link to the user's inbox.
6. The system displays a confirmation: "A recovery link has been sent to your email."
7. The student opens the link, enters a new password, and the system updates the database.

Alternate Flows

A1 — Unregistered Email:

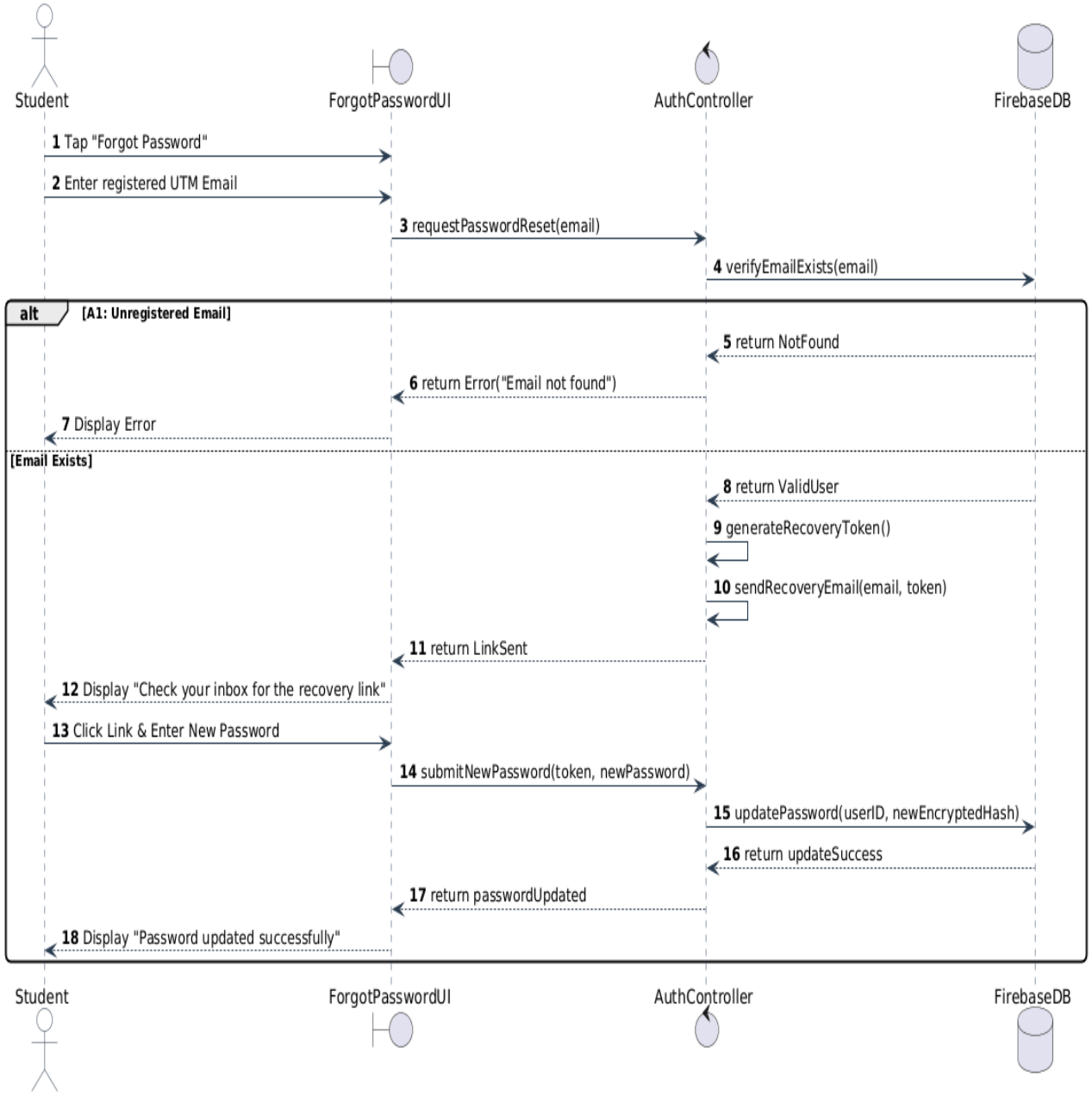
1. At step 4, the system queries the database and cannot find the provided email address.
2. The system displays an error message: "Email not found. Please check for typos or register a new account."

Post-conditions

1. The user's account password is safely updated in the database without compromising account security.

Table 3.2.5: Use Case Description of UC1103 — Recover Password

3.2.5.2. Sequence Diagram



3.3. Module 2: Profile & Trust Score

3.3.1. Structure of Module 2:

This module is designed to maintain the user's digital identity and community reputation. It implements a transparent 5-star Trust Score system to encourage honest behavior, ensuring that reliable users are rewarded with good ratings, while providing a formal channel to report severe non-compliance to university administrators.

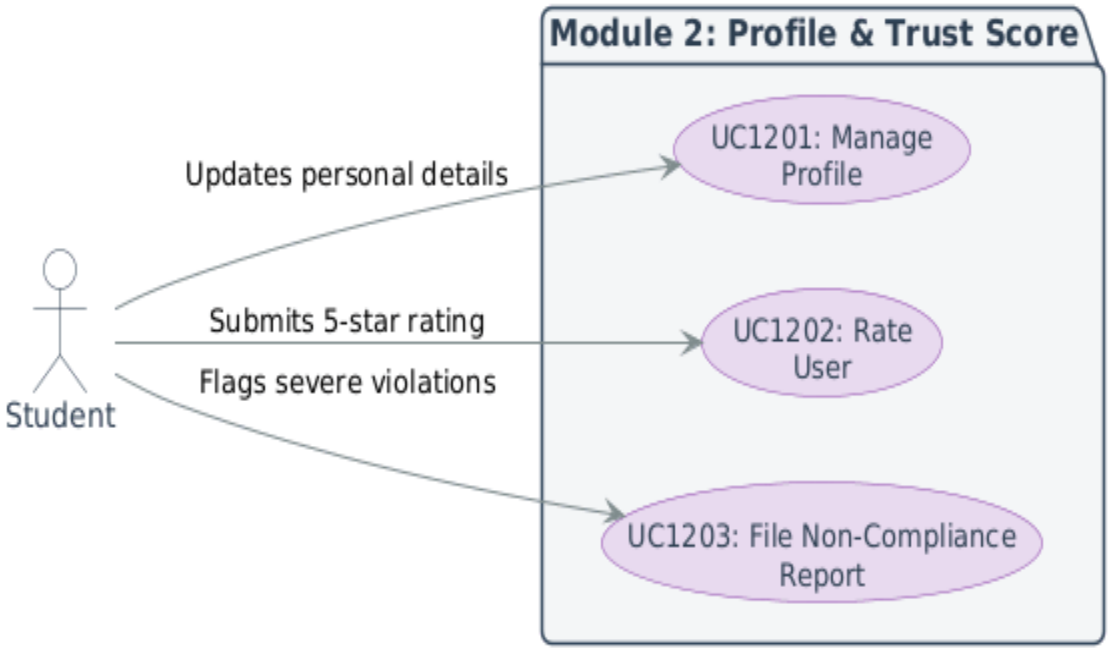


Figure 3.3: Package Diagram of Profile & Trust Score Module.

3.3.2. Specific Requirements

The Trust Score algorithm must dynamically recalculate the user's average rating immediately after every completed transaction. Users cannot rate themselves. Submitting a non-compliance report must immediately flag the offending user profile in the moderation database, requiring mandatory review by the platform administrator.

3.3.3. UC1201: Manage Profile

3.3.3.1. Use Case Specifications

Use Case: Manage Profile ID: UC1201	
Actors	Student Borrower, Student Lender, System
Module	Profile & Trust Score

Preconditions

1. The user is logged into their verified UTM Borrow account.

Flow of Events

1. The student navigates to the "My Profile" tab from the main navigation menu.
2. The system queries the database and displays the user's name, matric number, profile picture, and current Trust Score.
3. The student selects the "Edit Details" option.
4. The student updates their phone number and uploads a new profile picture.
5. The student taps "Save Changes".
6. The system validates the uploaded image format and saves the updated data to the database.
7. The system displays a success alert: "Profile details updated successfully.

Alternate Flows

A1 — Invalid Image Format:

1. At step 6, the system detects that the uploaded profile picture is an unsupported file type or exceeds the size limit.
2. The system blocks the save action and displays an error: "Please upload a valid JPEG or PNG image under 5MB."

Post-conditions

1. The user's profile details are globally updated and visible to other users on the platform.

Table 3.3.3: Use Case Description of UC1201 — Manage Profile

3.3.3.2. Sequence Diagram

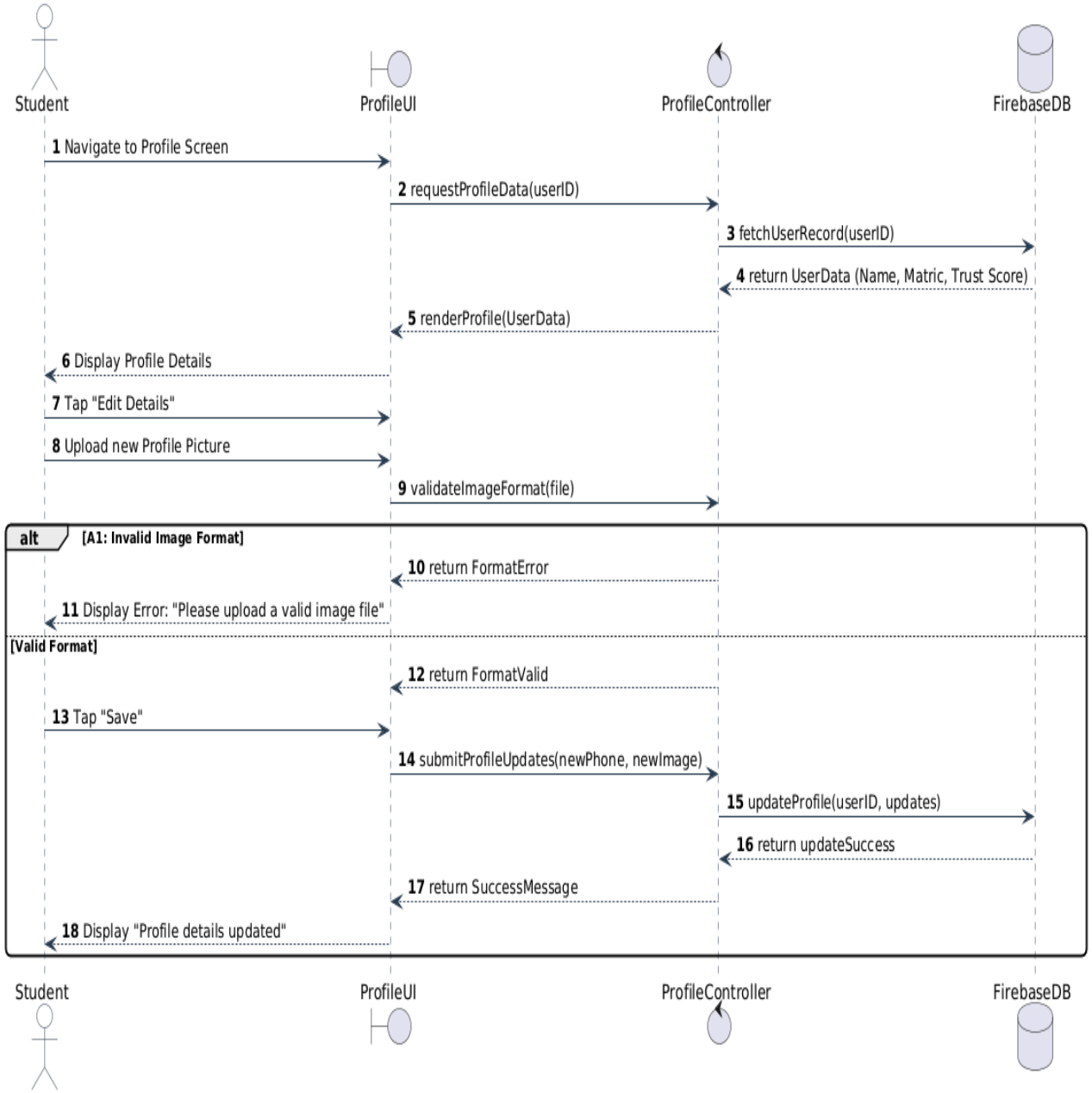


Figure 3.3.3: Sequence Diagram of UC1201 — Manage Profile

3.3.4. UC1202: Rate User

3.3.4.1. Use Case Specifications

Use Case: Rate User	ID: UC1202
Actors	Student Borrower, Student Lender, System
Module	Profile & Trust Score

Preconditions

1. A transaction has just been marked as "Completed" (the borrowed item was physically returned and scanned).

Flow of Events

1. The system automatically prompts the user with a dialog box asking them to rate their peer's behavior during the transaction.
2. The student selects a rating on a 1-to-5 star scale and optionally adds brief text feedback.
3. The student taps "Submit Rating".
4. The system calculates the new aggregate average Trust Score for the peer.
5. The system saves the new Trust Score and feedback to the peer's profile in the database.
6. The system displays a thank you message and closes the prompt.

Alternate Flows

A1 — User Dismisses Prompt:

1. At step 2, the user taps the "Dismiss" or "Remind Me Later" button.
2. The system closes the dialog and saves the rating task to the user's "Pending Actions" dashboard list to be completed later.

Post-conditions

1. The peer's public Trust Score is updated, influencing their future ability to borrow or lend items.

Table 3.3.4: Use Case Description of UC1202 — Rate User

3.3.4.2. Sequence Diagram

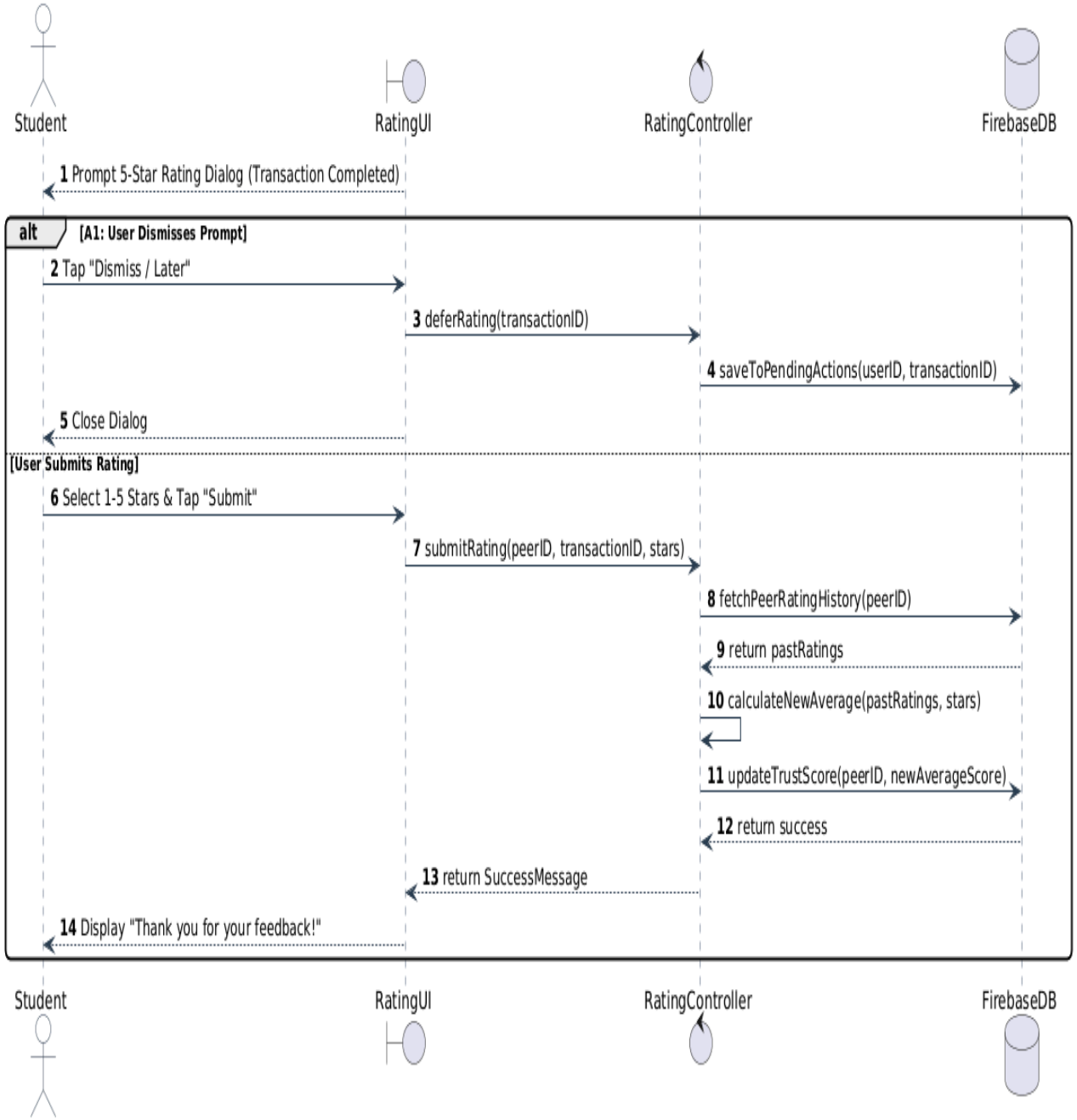


Figure 3.3.4: Sequence Diagram of UC1202 — Rate User

3.3.5. UC1203: Report User

3.3.5.1. Use Case Specifications

Use Case: Report User ID: UC1203	
Actors	Student Borrower, Student Lender, System
Module	Profile & Trust Score

Preconditions

1. The user has experienced a severe violation of platform trust (e.g., an unreturned item, severe physical damage, or harassment).

Flow of Events

1. The user navigates to the offending peer's public profile page.
2. The user taps the "Report User" button.
3. The system displays a structured reporting form prompting the user to select a violation category (e.g., Damaged Item, Scam) and input a detailed description.
4. The user completes the form and taps "Submit Report".
5. The system logs the report and immediately flags the offending user's account in the database.
6. The system routes the report to the Admin Moderation Queue for review.
7. The system displays a confirmation message: "Report submitted to UTM Administration.

Alternate Flows

A1 — Missing Description Details:

1. At step 4, the system detects that the text description field has been left blank.
2. The system prevents submission and highlights the field in red, displaying: "Please provide detailed information regarding the violation to proceed."

Post-conditions

1. A formal non-compliance report is lodged, and the offending user's profile is officially flagged for potential administrative disciplinary action.

Table 3.3.5: Use Case Description of UC1203 — Report User

3.3.5.2. Sequence Diagram

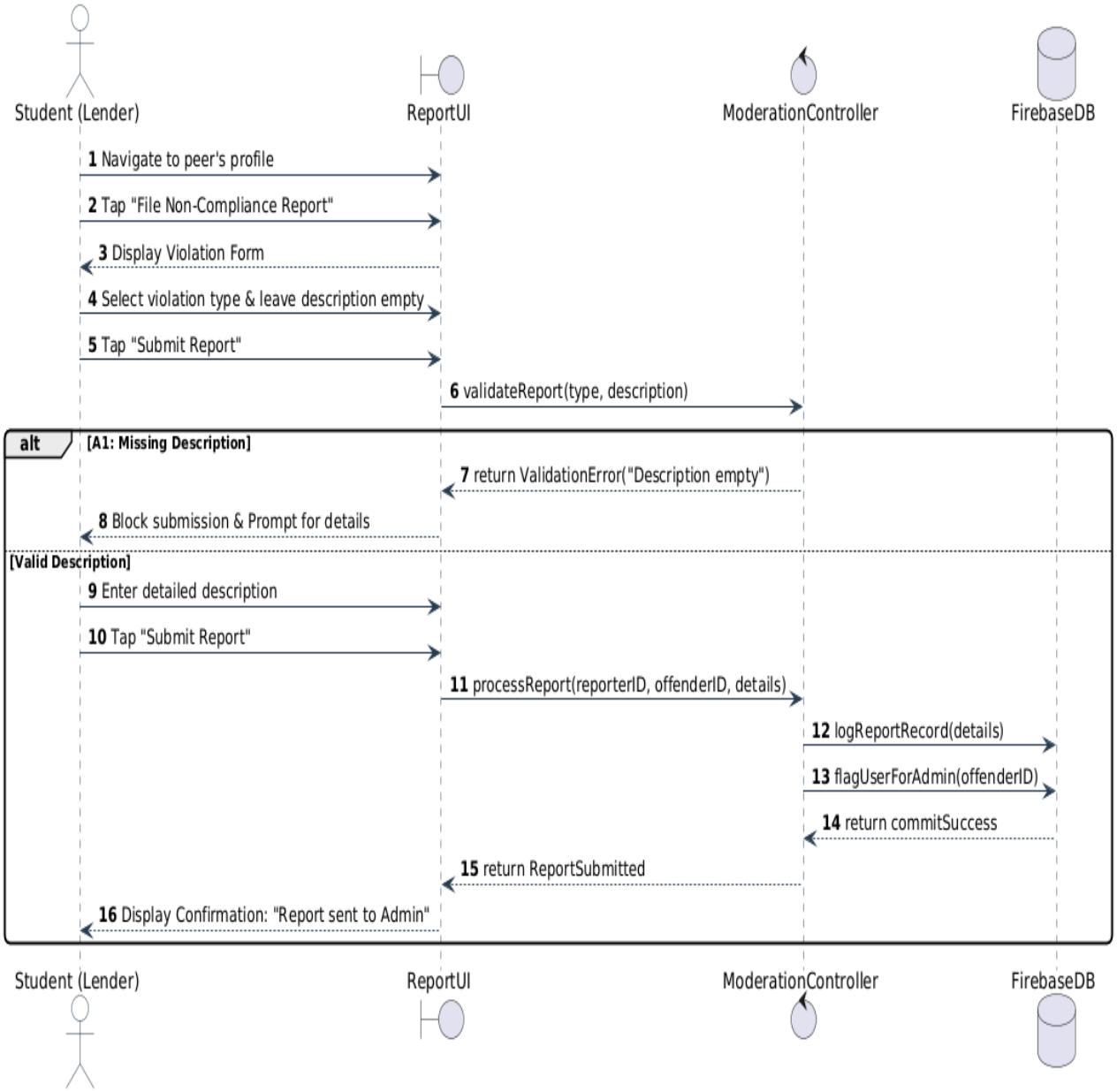
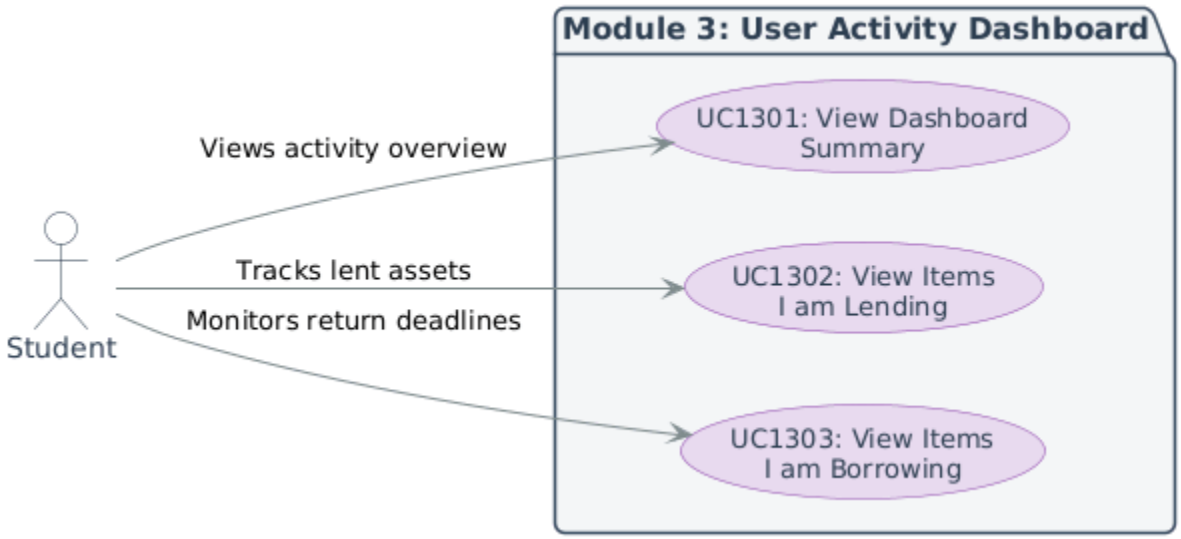


Figure 3.3.5: Sequence Diagram of UC1203 — Report User

3.4. Module 3: User Activity Dashboard

3.4.1. Structure of Module 3:

The User Activity Dashboard acts as the central tracking and organization hub for all user engagements on the platform. It neatly organizes real-time transaction data so users can clearly monitor what items they currently owe to others, and what items are owed to them, ensuring high accountability and timely returns.



3.4.2. Specific Requirements

The dashboard must fetch and synchronize real-time transaction data from the shared database. It must dynamically separate data into specific interface tabs depending on the user's current role (Lender vs. Borrower). The system must utilize color-coded visual alerts (e.g., Red) for items approaching their return deadlines.

3.4.3. UC1301: View Dashboard

3.4.3.1. Use Case Specifications

Use Case: View Dashboard ID: UC1301	
Actors	Student Borrower, Student Lender, System
Module	User Activity Dashboard
Preconditions	
1. The user is logged into the application.	
Flow of Events	
1. The student opens the application and taps the "Dashboard" icon on the main menu. 2. The system queries the database for all active and pending transactions linked to the user's specific ID. 3. The system calculates summary statistics based on the retrieved data. 4. The system displays a high-level overview featuring quick summary widgets (e.g., "Total Items Borrowed", "Pending Requests").	
Alternate Flows	
A1 — No Active Transactions Exist: 1. At step 3, the system queries the database and finds zero active records for the user. 2. The system renders an empty state UI displaying the message: "You have no active lending or borrowing commitments at this time."	
Post-conditions	
1. The user is presented with a clear, real-time overview of their platform activity and pending tasks.	

Table 3.4.3: Use Case Description of UC1301 — View Dashboard

3.4.3.2. Sequence Diagram

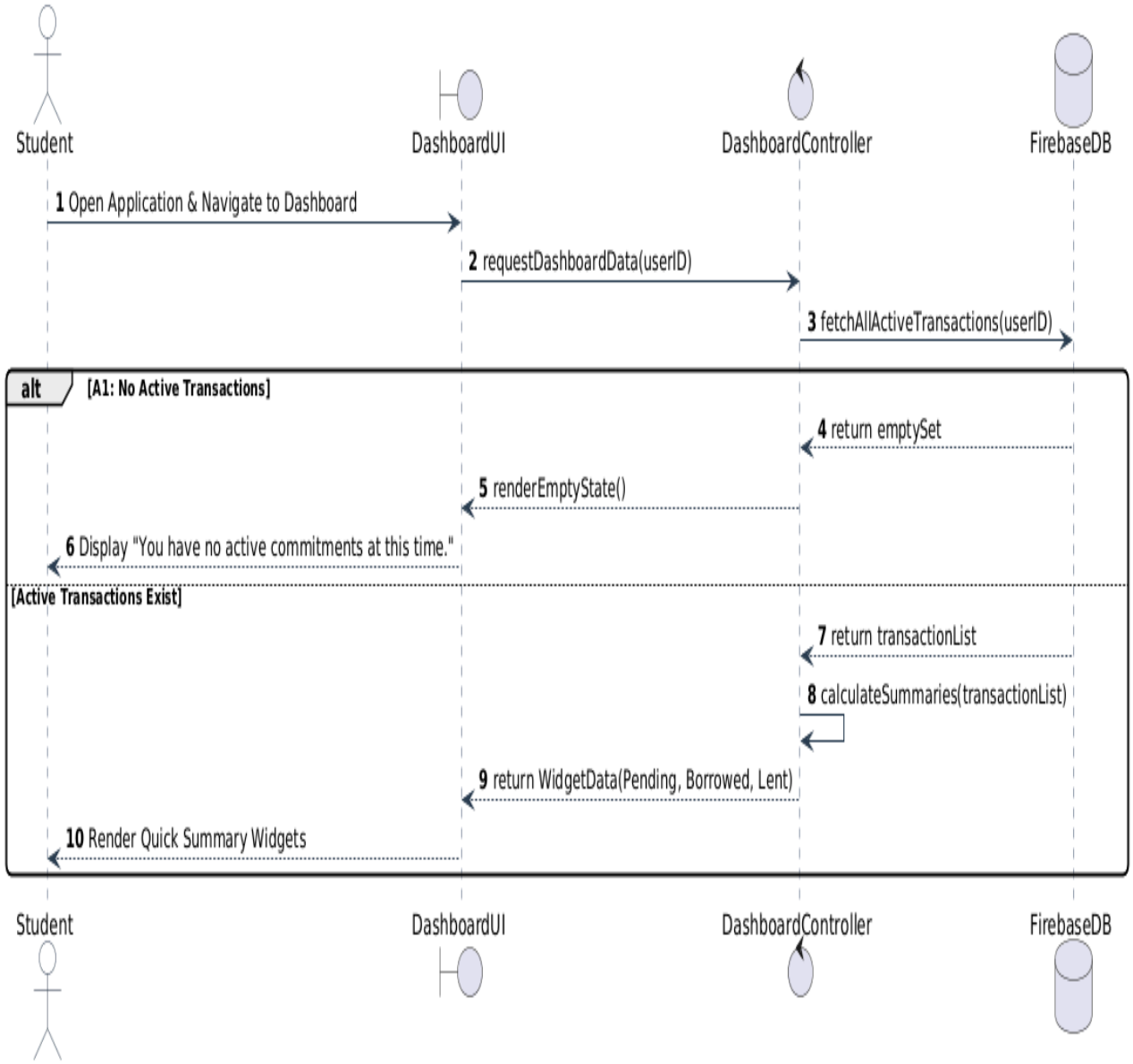


Figure 3.4.3: Sequence Diagram of UC1301 — View Dashboard

3.4.4. UC1302: View Lent Items

3.4.4.1. Use Case Specifications

Use Case: View Lent Items ID: UC1302	
Actors	Student Lender, System
Module	User Activity Dashboard

Preconditions

1. The user is currently acting as a lender for at least one active item on the platform.

Flow of Events

1. From the main Dashboard, the student taps the "Items I am Lending" tab.
2. The system filters the user's global transaction list to isolate only the records where the user is designated as the item owner.
3. The system retrieves the current status of each item (e.g., Pending Approval, Borrowed).
4. The system displays a detailed list showing the item name, the borrower's name, and the expected return date.

Alternate Flows

A1 — Tap Item for Full Details:

1. At step 4, the user taps on a specific item in the lending list.
2. The system navigates the user to a full transaction details screen displaying granular data, including QR code access.

Post-conditions

1. The lender can accurately track the exact physical location and status of their lent assets.

Table 3.4.4: Use Case Description of UC1302 — View Lent Items

3.4.4.2. Sequence Diagram

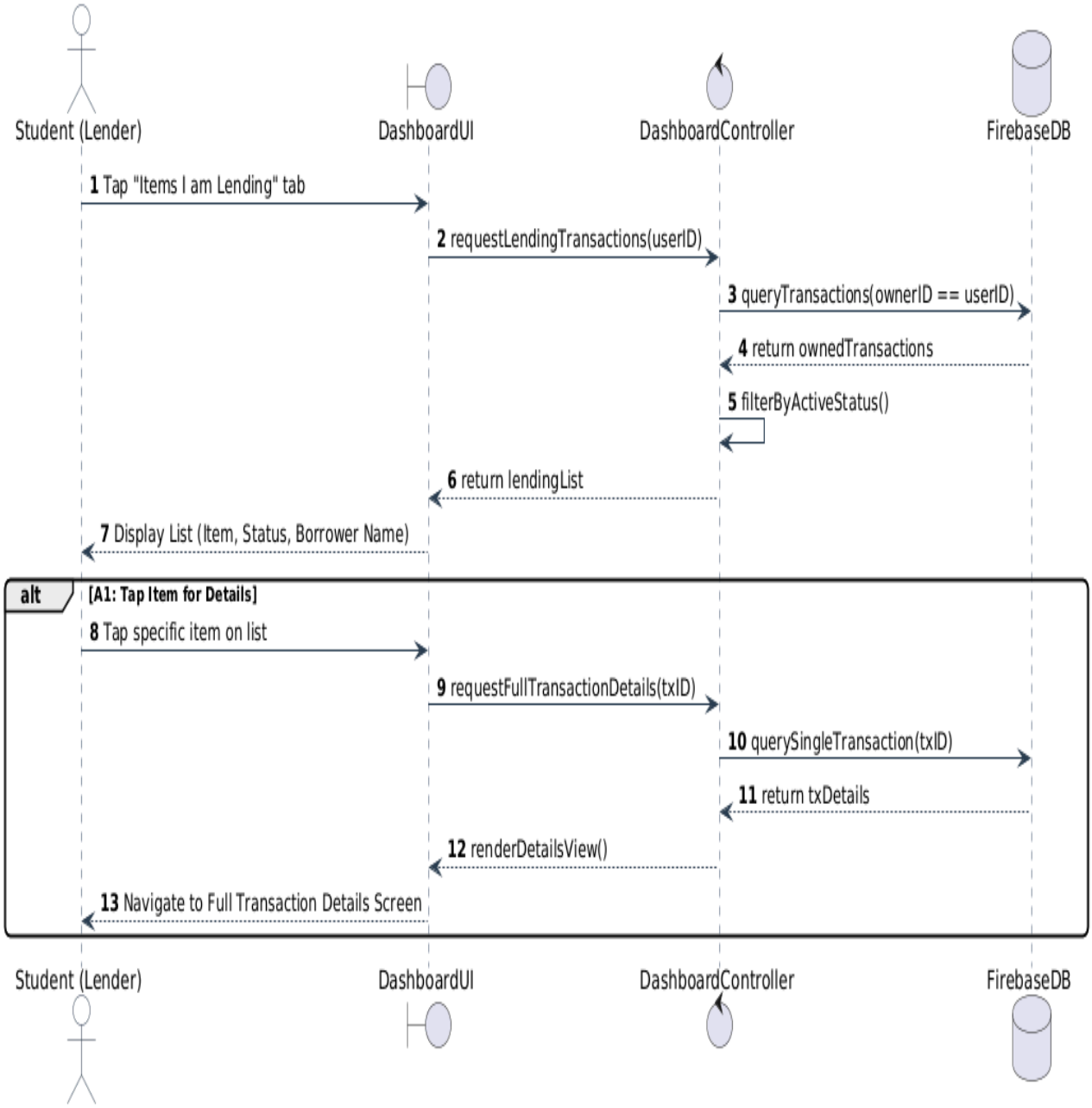


Figure 3.4.4: Sequence Diagram of UC1302 — View Lent Items

3.4.5. UC1303: View Borrowed Items

3.4.5.1. Use Case Specifications

Use Case: View Borrowed Items ID: UC1303	
Actors	Student Borrower, System
Module	User Activity Dashboard
Preconditions	
1. The user has initiated borrow requests or currently holds borrowed items in their possession.	
Flow of Events	
1. From the main Dashboard, the student taps the "Items I am Borrowing" tab. 2. The system filters the transaction database to show only the records where the user is designated as the borrower. 3. The system calculates the time remaining for each active lease based on the agreed-upon return dates. 4. The system displays the items in a chronological list prioritized by the closest return deadline.	
Alternate Flows	
A1 — Approaching Deadline Alert: 1. At step 3, the system detects that an item's return date is within 24 hours or has already passed. 2. The system automatically highlights the specific item listing in red and displays an "Urgent Return" badge to alert the borrower.	
Post-conditions	
1. The borrower is kept fully aware of their return responsibilities, significantly reducing the likelihood of late returns.	

Table 3.4.5: Use Case Description of UC1303 — View Borrowed Items

3.4.5.2. Sequence Diagram

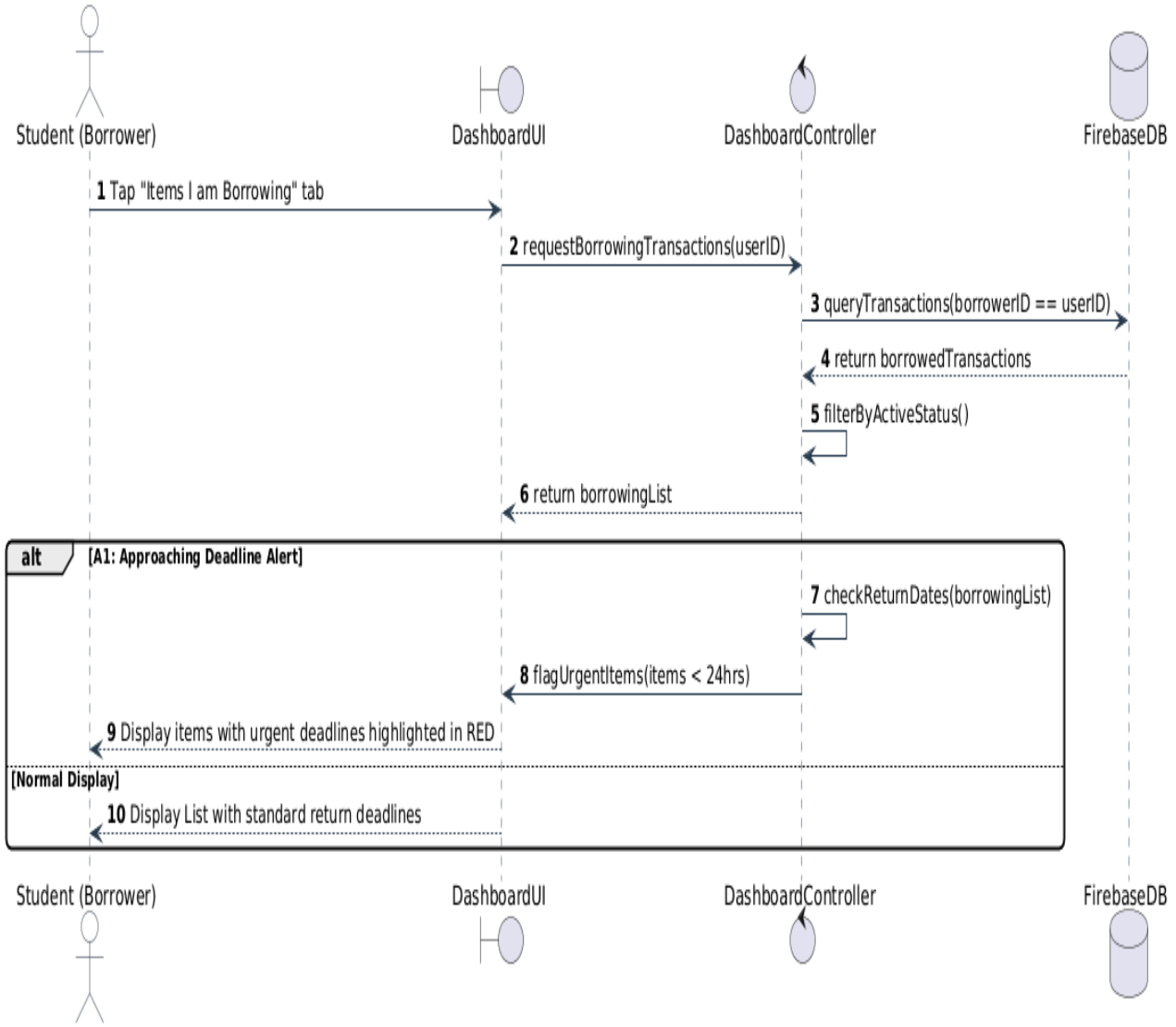


Figure 3.4.5: Sequence Diagram of UC1303 — View Borrowed Items

4. Resource Catalog & Discovery Subsystem - Muaz

4.1. Subsystem Overview

The Resource Catalog & Discovery Subsystem is the primary browsing and search engine of the UTM Borrow platform. It enables students and staff to discover available academic and extracurricular resources efficiently, view detailed metadata about each item, and filter listings based on relevant criteria such as category, condition, and physical proximity. This subsystem ensures that lenders' items gain visibility while providing borrowers with powerful discovery tools to locate exactly what they need without scrolling through unstructured chat logs.

4.1.1. Purpose

The primary purpose of this subsystem is to provide a structured, searchable, and filterable catalog of all available resources on campus. Unlike informal WhatsApp or Telegram groups where listings are buried in message history, this subsystem organizes items by category, condition, and location, allowing users to discover relevant resources in seconds. It also enables lenders to manage their own listings, ensuring that catalog data remains accurate, up-to-date, and trustworthy.

4.1.2. Problem Statement

Currently, students rely on unstructured social media groups to find items they need. Listings are easily lost in message threads, there is no standardized way to describe an item's condition or category, and there is no location-based discovery to find items near a specific residential college or faculty building. Borrowers waste time scrolling through irrelevant content, while lenders struggle to make their items visible to the right audience. The absence of structured metadata also makes it impossible to filter, sort, or compare items systematically.

4.1.3. Subsystem Objectives

Provide a Centralized Resource Catalog: Create a single, searchable repository of all available lendable items on campus, replacing fragmented WhatsApp/Telegram listings.

Enable Structured Item Management: Allow lenders to publish, edit, remove, and refresh their item listings with proper metadata including title, description, photos, category, condition, and location.

Implement Category & Condition Metadata: Standardize resource classification using predefined academic and lifestyle categories, along with condition ratings so borrowers know exactly what to expect.

Deliver Location-Based Discovery: Allow borrowers to filter items by proximity to their residential college or faculty, reducing travel time and encouraging hyper-local sharing.

4.1.4. SUBSYSTEM ARCHITECTURE OVERVIEW

4.1.4.1. Modules Structure

The Resource Catalog & Discovery Subsystem is organized into three integrated modules, each responsible for a distinct phase of resource discovery and management:

Module	Purpose & Scope
Item Listing & Management	Enables lenders to publish, edit, remove, and refresh their item listings. Includes photo upload, title/description editing, and automatic timestamping for "recently added" sorting.
Category & Condition Metadata	Provides structured dropdowns for item classification (e.g., Textbooks, Lab Equipment, Event Clothing) and condition rating (Like New, Good, Acceptable, Damaged).
Location-Based Filter	Allows borrowers to filter items by proximity to their residential college (e.g., Kolej Tuanku Canselor, Kolej Rahman Putra) or faculty building, with optional distance-based sorting.

4.1.4.2. Use Case Diagram

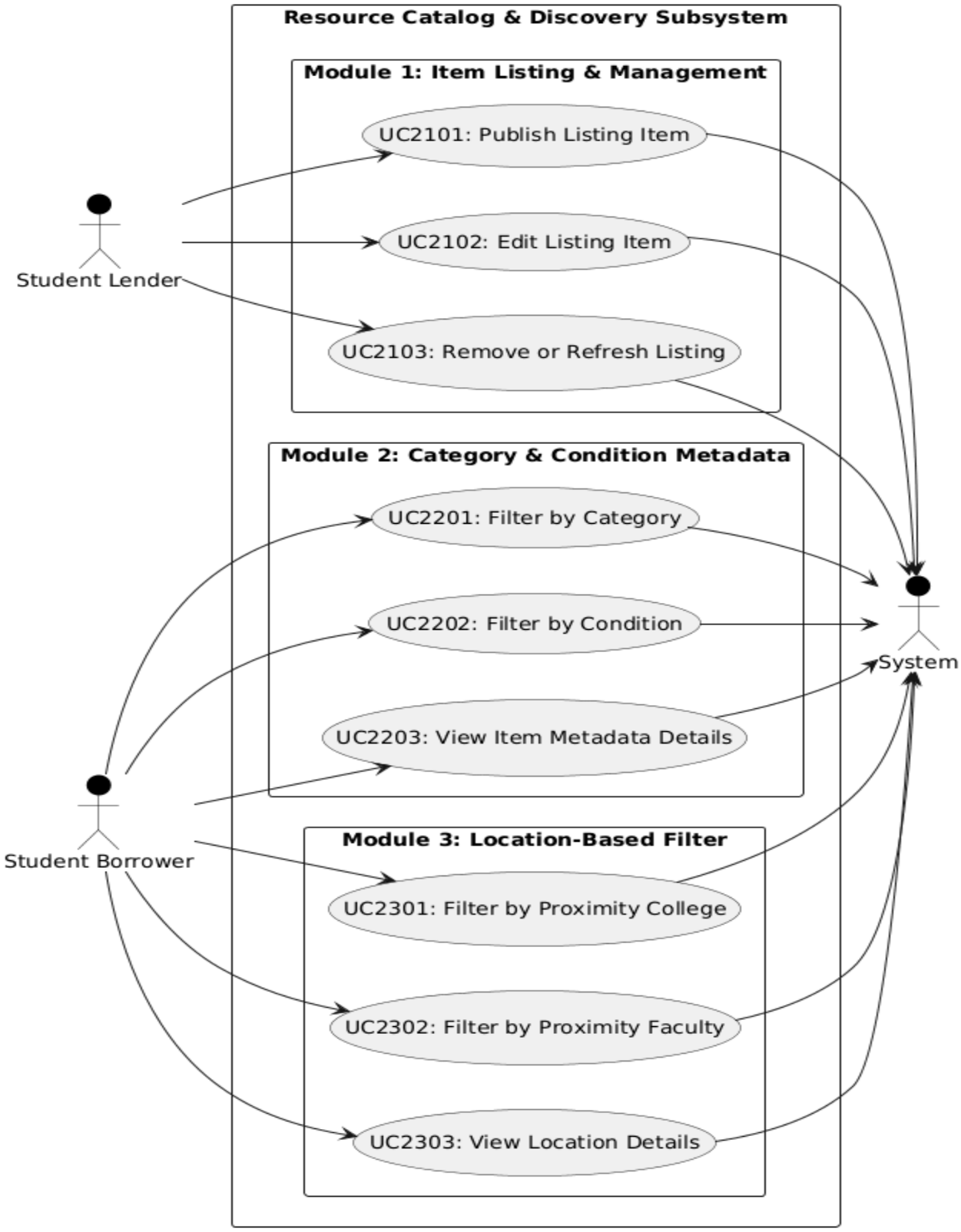


Figure 4.1.4: Use Case Diagram of the Resource Catalogue and Discovery Subsystem

4.2. Module 1: Item Listing & Management

4.2.1. Structure of Module 1:

This module enables lenders to publish and maintain their item listings. When a user wants to share a resource, they navigate to the "Add New Item" screen, upload photos, enter a title and description, select a category and condition, and optionally specify a location. Once published, the item becomes immediately visible in the Resource Catalog. Lenders can edit any field later, remove items that are no longer available, or "refresh" listings to bump them up in chronological sort order.

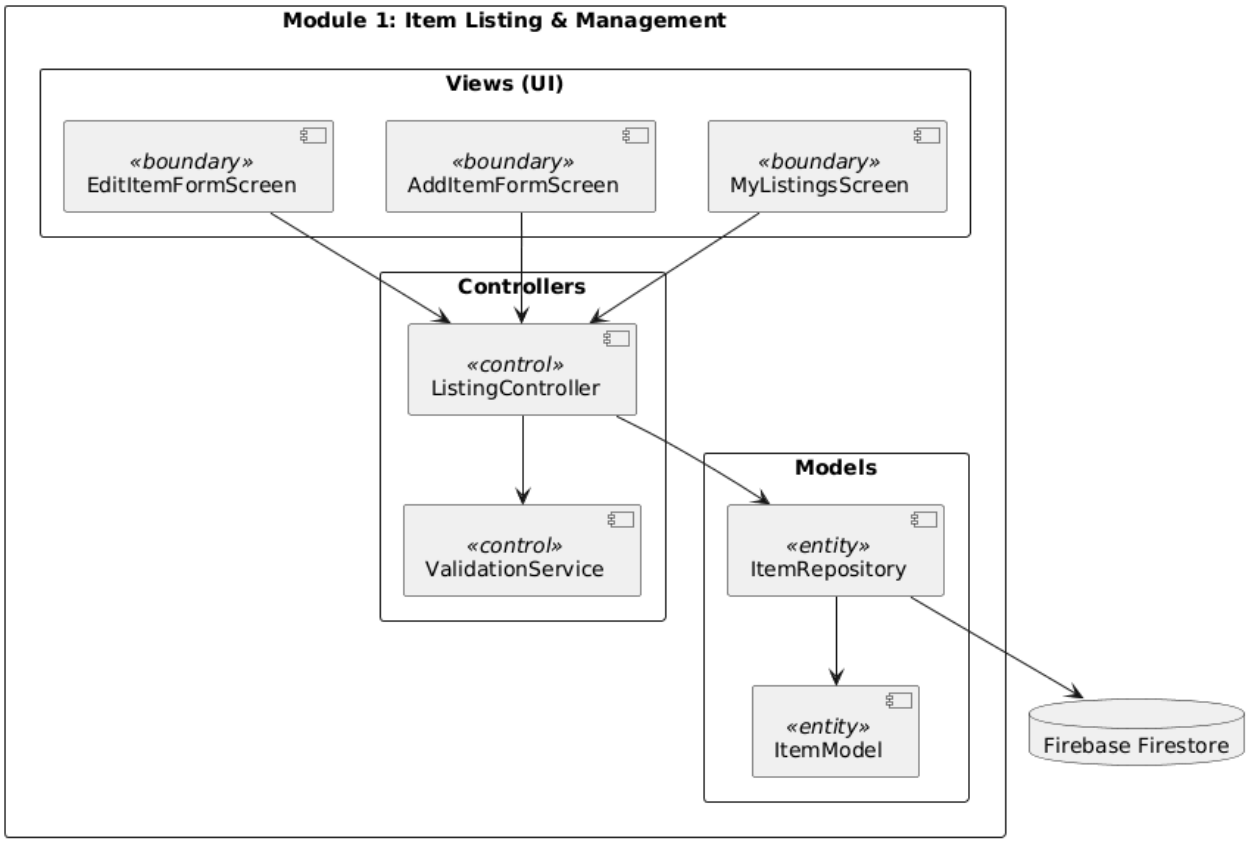


Figure 4.2.1: Package Diagram of the Item Listing and Management Module

4.2.2. Specific Requirements

All item listings must include at least one photo and a non-empty title. The system must prevent duplicate submission of the exact same item by the same lender within a 5-minute window. Upon successful publication, the item must appear immediately in the global catalog with an "Available" status. Lenders must be able to edit any field except the item ID. Removal must be irreversible (soft delete with audit flag). Refresh functionality must update a `last_refreshed_at` timestamp used for "Recently Added" sorting.

4.2.3. UC2101: Publish Listing Item

4.2.3.1. Use Case Specifications

Use Case: Publish Listing Item ID: UC2101	
Actors	Student Lender, System
Module	Item Listing and Management
Preconditions	
1. The lender is logged into their verified UTM Borrow account. 2. The lender has physical possession of the item they wish to list.	
Flow of Events	
1. The lender navigates to the "My Listings" tab and taps "Add New Item." 2. The system displays a publication form with fields: Title (required), Description, Photos (minimum 1), Category (dropdown), Condition (dropdown), Location (optional). 3. The lender enters all required information and taps "Publish." 4. The system validates that Title is not empty and at least one photo is attached. 5. The system writes the new item record to the database with status = "Available," timestamps = current UTC time, and lender_id = logged-in user ID. 6. The system generates a unique item ID (UUID v4). 7. The system displays a success message: "Item listed successfully! It is now visible in the Resource Catalog." 8. The system navigates the lender back to "My Listings."	
Alternate Flows	
A1: Missing Title: 1. At step 4, the system detects an empty title field. The system displays: "Please enter a title for your item." The lender must provide a title before publishing. **A2:** No Photo Uploaded: 1. At step 4, the system detects zero photos attached. The system displays: "Please upload at least one photo of your item." The lender must upload a photo. **A3:** Duplicate Submission: 1. At step 5, the system detects an identical item (same title, same lender, within 5 minutes). The system displays: "You already listed this item recently. Please wait before listing it again."	
Post-conditions	
1. A new item record is created in the database with "Available" status. 2. The item appears in the Resource Catalog for all borrowers. 3. The item appears in the lender's "My Listings" dashboard.	

Table 4.2.3: Use Case Description of UC2101 — Publish Listing Item

4.2.3.2.

4.2.3.3. Sequence Diagram

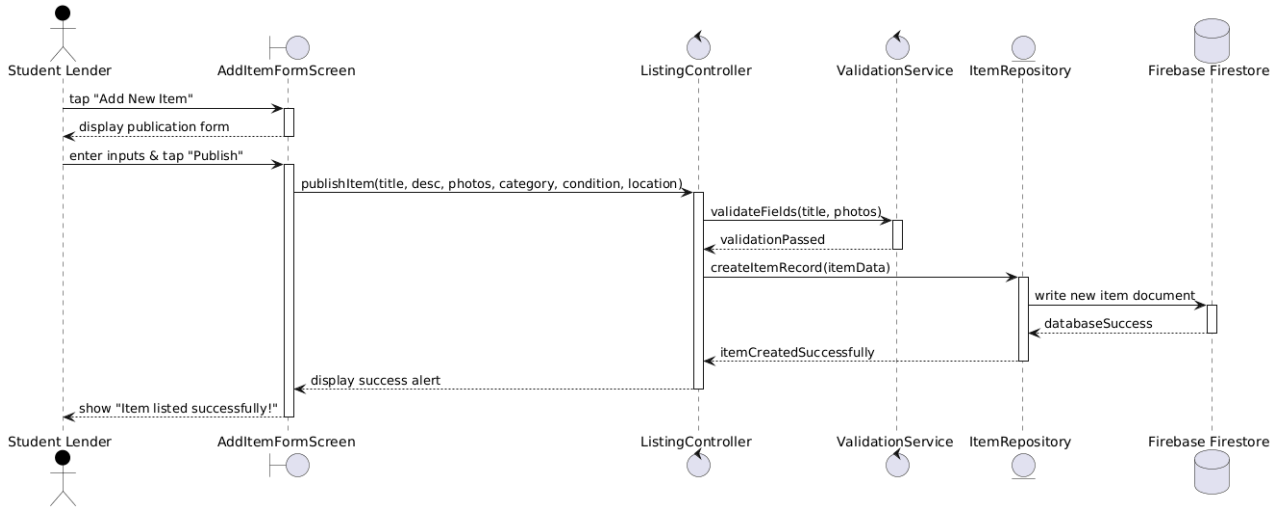


Figure 4.2.3: Sequence Diagram of UC2101 — Publish Listing Item

4.2.4. UC2102: Edit Item Listing

4.2.4.1. Use Case Specifications

Use Case: Edit Item Listing ID: UC2102	
Actors	Student Lender, System
Module	Item Listing and Management

Preconditions

1. The lender is logged into their verified UTM Borrow account.
2. The lender has at least one existing item listed in "My Listings."
3. The item is not currently in "Borrowed" status (can edit, but with warning).

Flow of Events

1. The lender navigates to "My Listings" and selects an item they wish to edit.
2. The system displays the full item details screen.
3. The lender taps the "Edit" button.
4. The system displays an editable form pre-populated with existing values (title, description, photos, category, condition, location).
5. The lender modifies one or more fields and taps "Save Changes."
6. The system validates that the Title field is not empty and at least one photo remains.
7. he system updates the database with the new values and refreshes the updated_at timestamp.
8. he system displays a success message: "Item updated successfully."
9. The system returns to the item details screen showing the updated information.

Alternate Flows
A1: Item is Currently Borrowed: **1.** At step 3, the system detects item status = "Borrowed." The system displays a warning: "This item is currently borrowed. Editing will not affect the active transaction. Continue?" Lender confirms to proceed. **A2:** Lender Removes All Photos: **1.** At step 6, the system detects zero photos remaining. The system displays: "Your item must have at least one photo." The lender must keep or re-add a photo.
Post-conditions
1. The item listing's metadata is updated in the database. **2.** The updated information is immediately reflected in the Resource Catalog.

Table 4.2.4: Use Case Description of UC2102 — Edit Item Listing

4.2.4.2.

4.2.4.3. Sequence Diagram

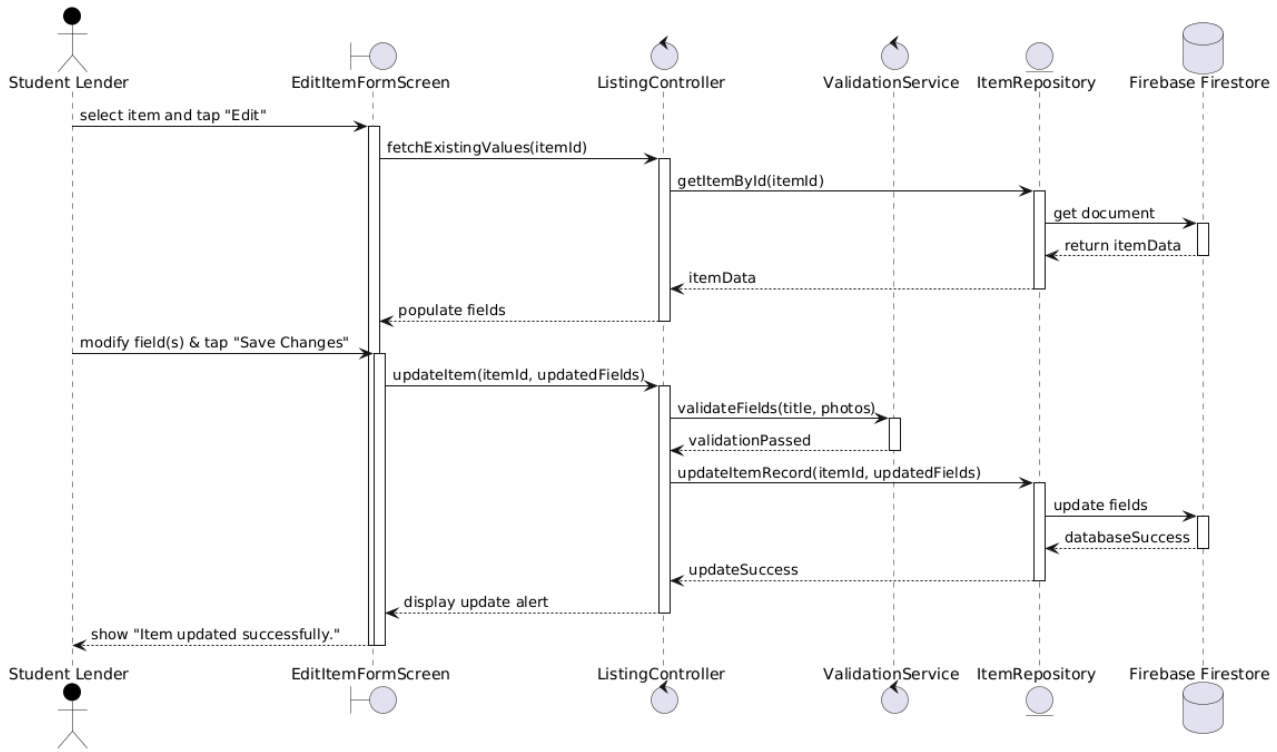


Figure 4.2.4: Sequence Diagram of UC2102 — Edit Listing Item

4.2.5. UC2103: Remove or Refresh Listing

4.2.5.1. Use Case Specifications

Use Case: Remove or Refresh Listing ID: UC2103	
Actors	Student Lender, System
Module	Item Listing and Management

Preconditions

1. The lender is logged into their verified UTM Borrow account.
2. The lender has at least one existing item listed in "My Listings."

Flow of Events (Remove)

1. Lender navigates to "My Listings" and selects an item.
2. Lender taps "Remove Listing."
3. System displays confirmation: "Are you sure? This action cannot be undone."
4. Lender confirms.
5. System updates item status to "Removed" (soft delete) and hides it from the catalog.
6. System displays success message: "Item removed from catalog."

Flow of Events (Refresh)

1. Lender navigates to "My Listings" and selects an item.
2. Lender taps "Refresh Listing."
3. System updates the last_refreshed_at timestamp to the current UTC time.
4. System displays success message: "Listing refreshed! It will appear higher in 'Recently Added' sort."

Alternate Flows

A1: Remove Active Borrowed Item:

1. At step 3 for Remove, the system detects status = "Borrowed." System displays: "This item is currently borrowed. Removing it will not cancel the transaction. Continue?" Lender confirms.

Post-conditions (Remove)

1. Item is hidden from the Resource Catalog.
2. Item record remains in the database with status = "Removed" for audit purposes.

Post-conditions (Refresh)

1. last_refreshed_at timestamp is updated.
2. Item moves to the top when sorting by "Recently Added."

Table 4.2.5: Use Case Description of UC2103 — Remove or Refresh Listing

4.2.5.2.

4.2.5.3. Sequence Diagram

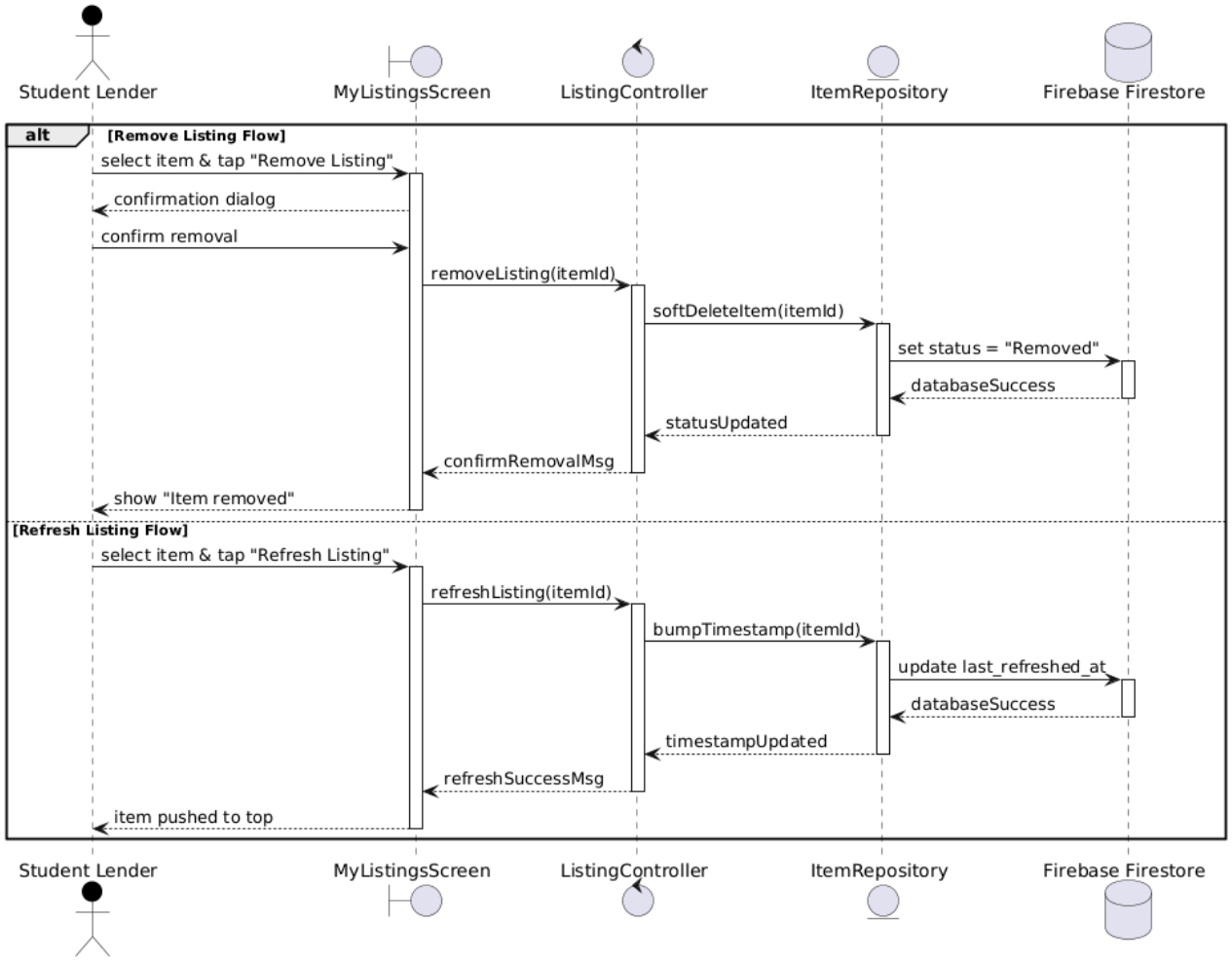


Figure 4.2.5: Sequence Diagram of UC2103 — Remove or Refresh Listing

4.3. Module 2: Category & Condition Metadata

4.3.1. Structure of Module 2:

This module provides standardized classification for all items in the catalog. Instead of free-text categories (which lead to inconsistency), lenders must select from a predefined list of academic and lifestyle categories. Similarly, every item must have a condition rating so borrowers know what to expect. The system uses these metadata fields to power faceted filtering and sorting on the catalog screen.

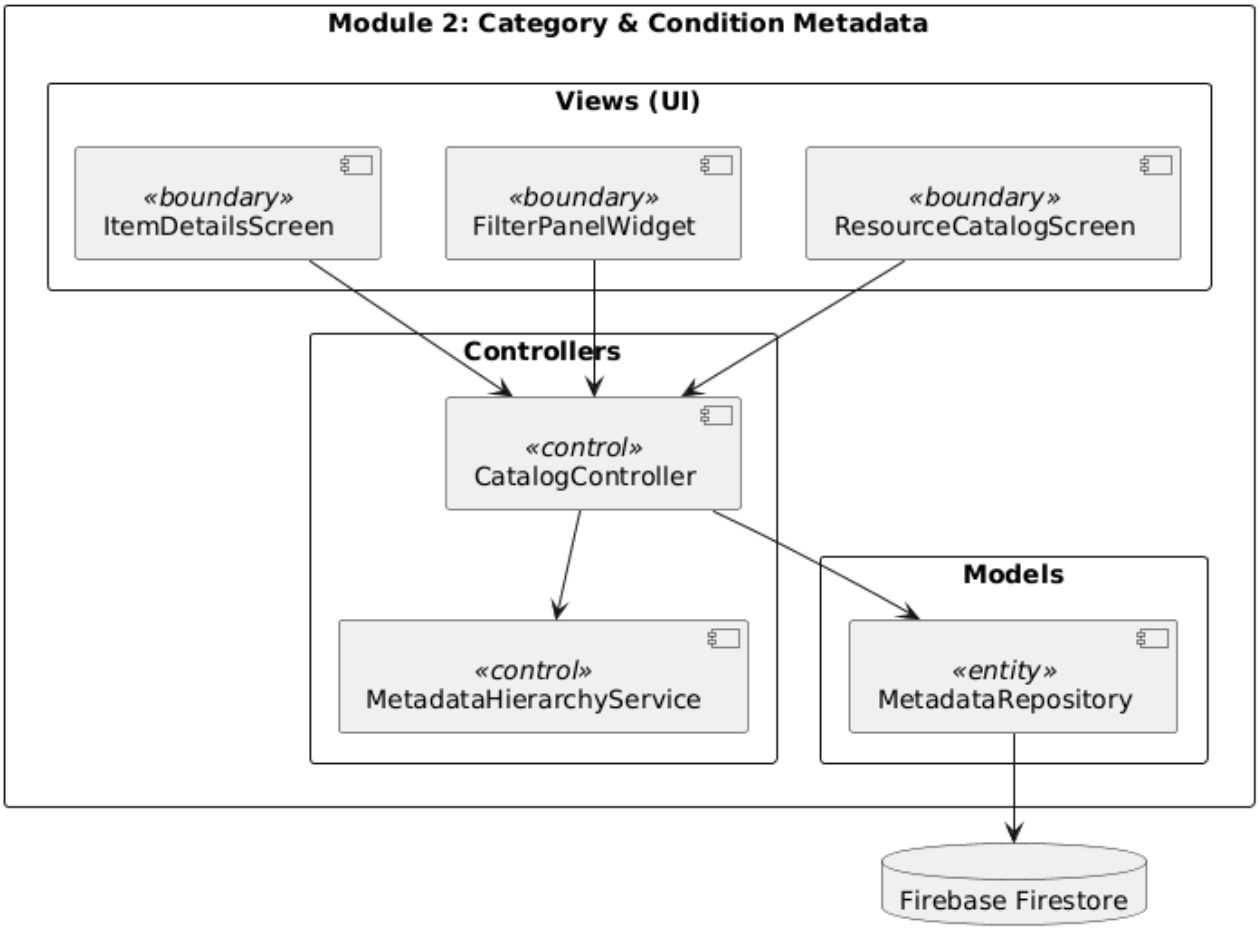


Figure 4.3.1: Package Diagram of the Category and Condition Metadata Module

4.3.2. Specific Requirements

The category dropdown must contain at least the following options: Textbooks, Lab Equipment, Engineering Tools, Event Clothing, Electronics, Sports Equipment, Stationery, Other. The condition dropdown must contain: Like New (used once or twice), Good (normal wear), Acceptable (functional but visible wear), Damaged (use with caution — must disclose damage in description). Categories and conditions must be configurable by an administrator in future versions. Filters must allow multi-select for categories. The condition filter must be single-select (Lowest acceptable condition).

4.3.3. UC2201: Filter by Category

4.3.3.1. Use Case Specifications

Use Case: Filter by Category ID: UC2201	
Actors	Student Borrower, System
Module	Category and Condition Metadata

Preconditions

1. The borrower is logged in
2. The Resource Catalog screen is open.

Flow of Events

1. Borrower taps the "Filter" icon on the catalog screen.
2. System displays filter panel with Category section showing all available categories as checkboxes.
3. Borrower selects one or more categories (e.g., "Textbooks" + "Lab Equipment").
4. Borrower taps "Apply Filters."
5. System queries the database for all available items where category IN (selected list) AND status = "Available."
6. The system displays the filtered result list.
7. The system shows a count of matching results.

Alternate Flows

A1: No Results Found:

1. At step 6, the query returns zero items. System displays: "No items found in the selected categories. Try removing some filters."

A2: Borrower Clears Filters:

1. Borrower taps "Clear All" in the filter panel. System removes all category selections and displays the full catalog.

Post-conditions

1. Catalog displays only items matching the selected categories.
2. The active filter indicator is displayed on the catalog screen.

Table 4.3.3: Use Case Description of UC2201 — Filter by Category

4.3.3.2.

4.3.3.3. Sequence Diagram

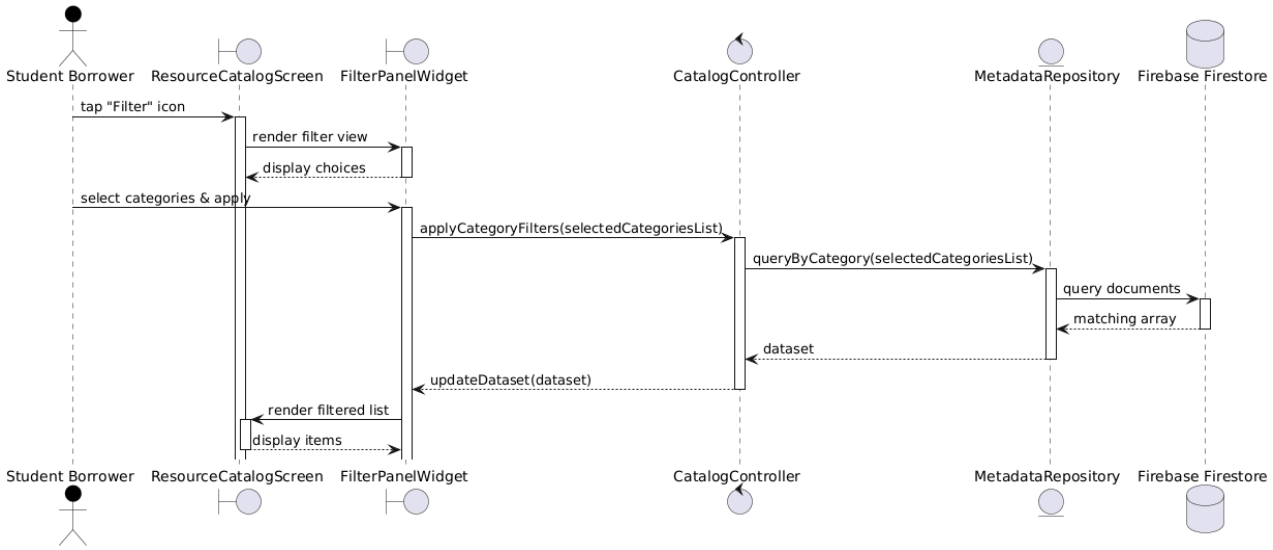


Figure 4.3.3: Sequence Diagram of UC2201 — Filter by Category

4.3.4. UC2202: Filter by Condition

4.3.4.1. Use Case Specifications

Use Case: Filter by Condition ID: UC2202	
Actors	Student Borrower, System
Module	Category and Condition Metadata

Preconditions

1. The borrower is logged in
2. The Resource Catalog screen is open.

Flow of Events

1. Borrower taps the “Filter” icon.
2. System displays filter panel with Condition section showing radio buttons: Any, Like New, Good, Acceptable, Damaged.
3. Borrower selects a minimum acceptable condition (e.g., “Good” means show Like New + Good, but not Acceptable or Damaged).
4. Borrower taps “Apply Filters.”
5. System calculates condition hierarchy: Any = all; Good = Like New + Good; Acceptable = Like New + Good + Acceptable; Damaged = all.
6. System queries database for items where `condition_rating` >= selected threshold.
7. System displays filtered results.

Alternate Flows

A1: Borrower Selects “Any”:

1. System does not apply condition filtering.

A2: No Results:

1. System displays "No items match your condition requirements."

Post-conditions

1. Catalog shows only items meeting the minimum condition threshold.

Table 4.3.4: Use Case Description of UC2202 — Filter by Condition

4.3.4.2.

4.3.4.3. Sequence Diagram

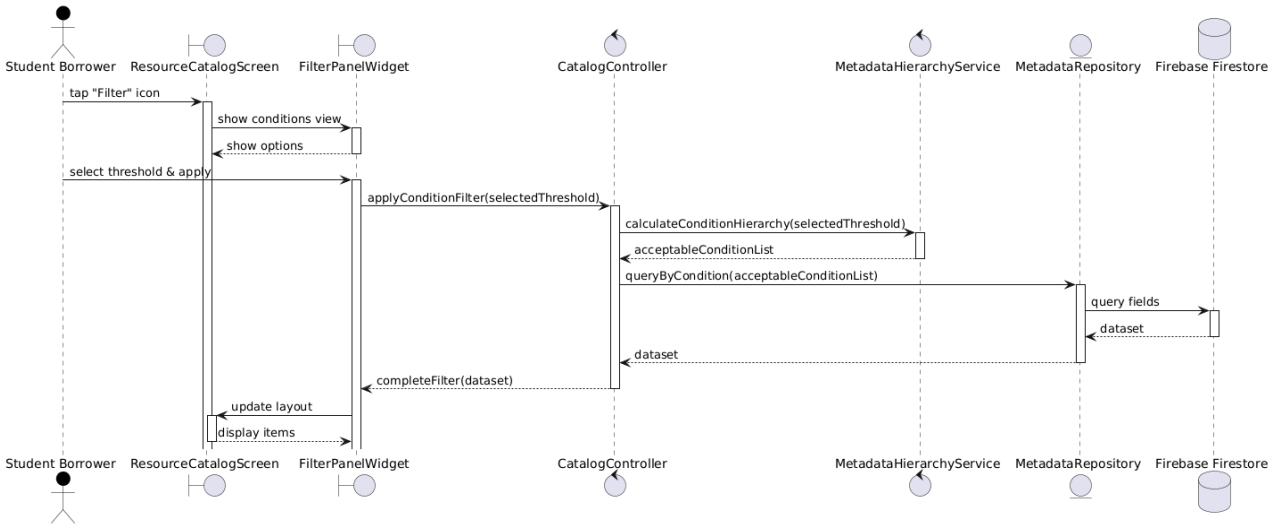


Figure 4.3.4: Sequence Diagram of UC2202 — Filter by Category

4.3.5. UC2203: Combined Category + Condition Search

4.3.5.1. Use Case Specifications

Use Case: Combined Category + Condition Search ID: UC2203	
Actors	Student Borrower, System
Module	Category and Condition Metadata

Preconditions

1. The borrower is logged in
2. The Resource Catalog screen is open.

Flow of Events

1. Borrower opens the filter panel.
2. Borrower selects multiple categories (e.g., "Textbooks" and "Electronics").
3. Borrower selects minimum condition = "Good."
4. Borrower taps "Apply Filters."
5. System constructs a compound query: category IN (list) AND condition_rating >= threshold AND status = "Available".

6. System executes the query against the database.
7. The system displays the combined filtered results.
8. System shows the active filter summary: “Categories: Textbooks, Electronics

Alternate Flows

A1: No Results:

1. System displays “No items match your filters. Try expanding categories or lowering the condition requirement.”

Post-conditions

1. Catalog displays items that satisfy both category and condition criteria simultaneously.
2. Both filter indicators are active.

Table 4.3.5: Use Case Description of UC2203 — Combined Category + Condition Search

4.3.5.2.

4.3.5.3. Sequence Diagram

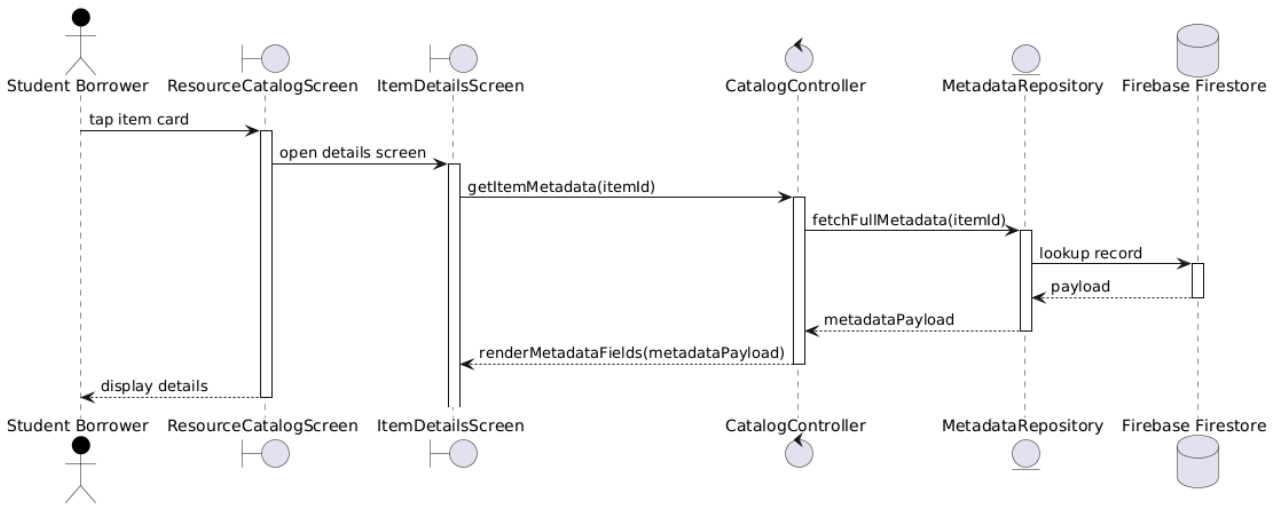


Figure 4.3.5: Sequence Diagram of UC2203 — Combined Category + Condition Search

4.4. Module 3: Location-Based Filter

4.4.1. Structure of Module 3:

This module allows borrowers to discover items near their physical location on campus. Borrowers can either manually select their residential college or faculty from a dropdown, or grant location permissions for automatic proximity detection. The system then filters or sorts items by distance, prioritizing resources that are close by to reduce travel friction and encourage hyper-local sharing.

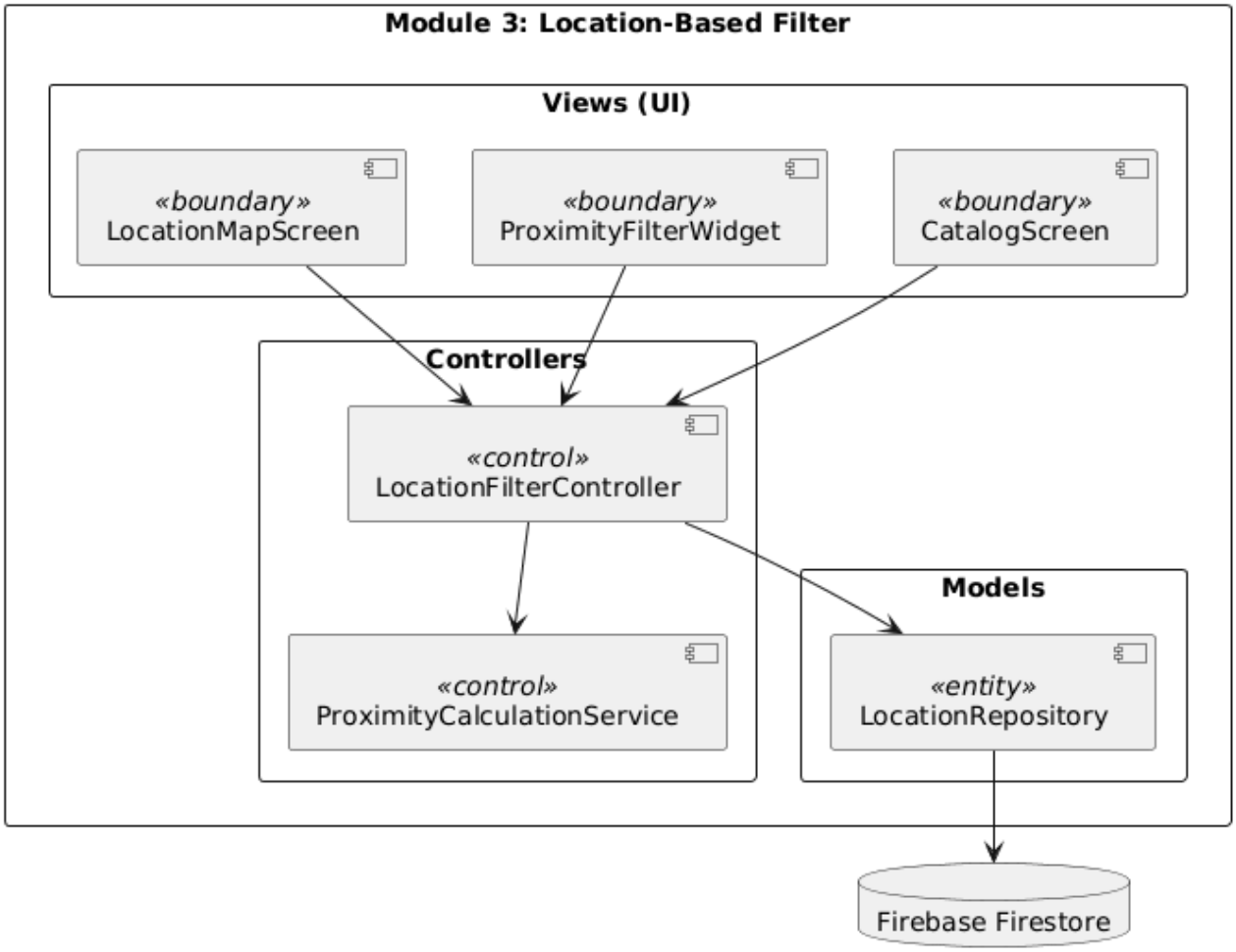


Figure 4.4.1: Package Diagram of the Location Based Filter Module

4.4.2. Specific Requirements

Location data must be optional; users without location permissions can still use manual selection. The location dropdown must include all UTM residential colleges (e.g., Kolej Tuanku Canselor, Kolej Rahman Putra, Kolej Tun Hussein Onn, etc.) and major faculties. Distance calculation must use approximate campus coordinate mapping (predefined for each college/faculty). If automatic location is used, the system must request permission only once and handle denial gracefully. Sorting by distance must be ascending (closest first).

4.4.3. UC2301: Filter by Proximity (Manual College/ Faculty Selection)

4.4.3.1. Use Case Specifications

Use Case: Filter by Proximity (Manual College/ Faculty Selection) ID: UC2301	
Actors	Student Borrower, System
Module	Location Based Filter

Preconditions

1. The borrower is logged in
2. The Resource Catalog screen is open.

Flow of Events

1. Borrower taps the "Location Filter" icon.
2. The system displays a dropdown of all UTM residential colleges and faculties.
3. Borrowers select their location (e.g., "Kolej Tuanku Canselor").
4. System maps the selected location to a set of predefined campus coordinates.
5. The system queries the database for all available items, calculates approximate distance for each based on the lender's listed location or default radius, and returns items within a 2km radius by default.
6. The system displays results sorted by distance (closest first).
7. System shows a note: "Showing items near Kolej Tuanku Canselor."

Alternate Flows

A1: No Items Near Selected Location:

1. System displays: "No items found near your selected location. Try expanding the distance or removing the location filter."

A2: Lender Did Not Specify Location:

1. Items without a lender-provided location are shown at the end of the list with a note "Location not specified."

Post-conditions

1. Catalog shows items prioritized by physical proximity to the borrower's selected college/faculty.

Table 4.4.3: Use Case Description of UC2301 — Filter by Proximity (Manual College/ Faculty Selection)

4.4.3.2.

4.4.3.3. Sequence Diagram

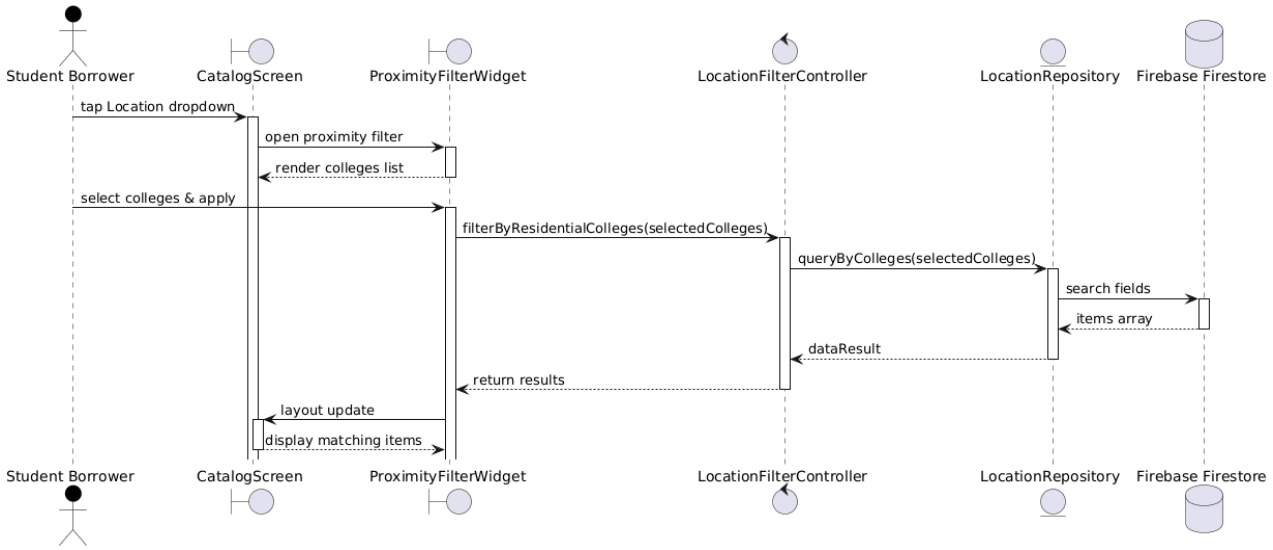


Figure 4.4.3: Sequence Diagram of UC2301 — Filter by Proximity (Manual College/ Faculty Selection)

4.4.4. UC2302: Automatic Location Detection (GPS)

4.4.4.1. Use Case Specifications

Use Case: Automatic Location Detection (GPS) ID: UC2302	
Actors	Student Borrower, System
Module	Location Based Filter
Preconditions	
1. The borrower is logged in **2.** Borrower's device has GPS/location services enabled. **3.** The application has permission to access location.	
Flow of Events	
1. Borrower taps the "Location Filter" icon. **2.** Borrower taps "Use My Current Location." **3.** System prompts: "Allow UTM Borrow to access your location?" (if not already granted). **4.** Borrower grants permission. **5.** The system fetches the device's current GPS coordinates. **6.** System queries the database for available items within a 2km radius of those coordinates. **7.** The system displays results sorted by distance. **8.** System shows: "Showing items near your current location."	
Alternate Flows	
A1: Permission Denied:	

1. At step 4, the borrower denies permission. System displays: “Location access denied. You can manually select your college instead.” System falls back to manual selection dropdown.

A2: GPS Unavailable:

1. Items without a lender-provided location are shown at the end of the list with a note “Location not specified.”

Post-conditions

1. Catalog shows items sorted by proximity to the borrower's real-time GPS location.

2. Location permission is stored for future sessions (user can revoke later).

Table 4.4.4: Use Case Description of UC2302 — Automatic Location Detection (GPS)

4.4.4.2.

4.4.4.3. Sequence Diagram

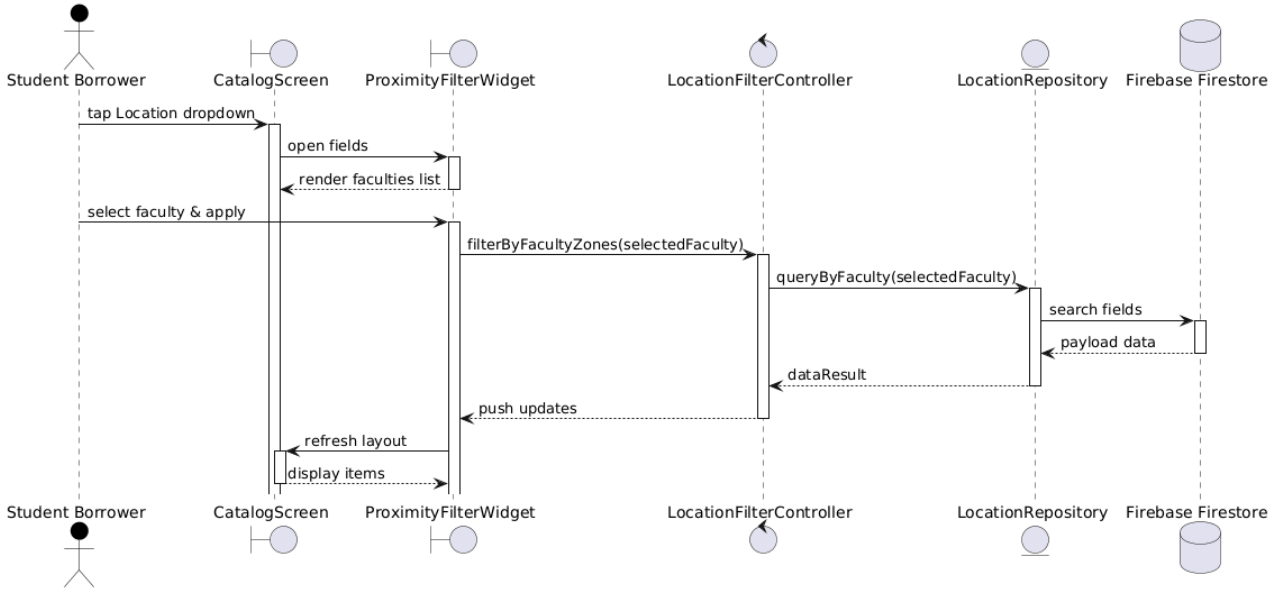


Figure 4.4.4: Sequence Diagram of UC2302 — Automatic Location Detection (GPS)

4.4.5. UC2303: Sort by Distance (Ascending/ Descending)

4.4.5.1. Use Case Specifications

Use Case: Sort by Distance (Ascending/ Descending) ID: UC2303	
Actors	Student Borrower, System
Module	Location Based Filter

Preconditions

1. 1. Borrower has already applied a location filter (manual or automatic).

2. At least one item exists within the search radius.

Flow of Events

1. On the filtered catalog results screen, the borrower taps the “Sort” dropdown.
2. The system displays sort options: “Distance: Nearest First” and “Distance: Farthest First.”
3. Borrower selects “Distance: Nearest First.”
4. The system reorders the existing result set in ascending order by calculated distance.
5. System refreshes the catalog display with the new order.

Alternate Flows

A1: No Location Filter Applied:

1. Borrower attempts to sort by distance without any location filter. System displays: “Please select a location first (manual or GPS) before sorting by distance.”

Post-conditions

1. The catalog list is reordered according to the selected distance sort option.

2. The active sort indicator is displayed.

Table 4.4.5: Use Case Description of UC2303 — Sort by Distance (Ascending/ Descending)

4.4.5.2.

4.4.5.3. Sequence Diagram

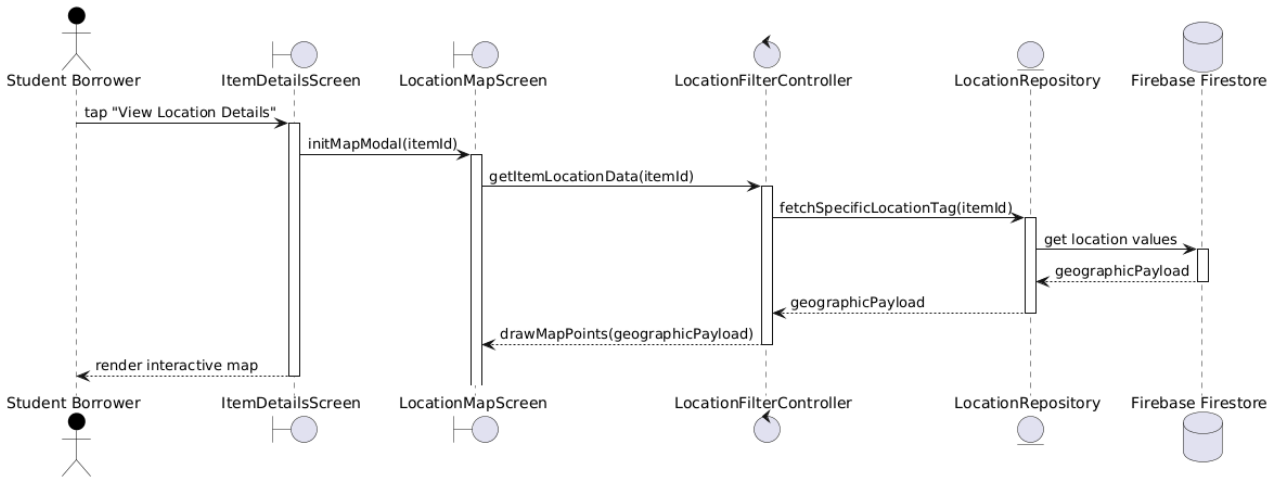


Figure 4.4.5: Sequence Diagram of UC2303 — Sort by Distance (Ascending/ Descending)

5. Transaction & Handover Subsystem - Alsakkaf

5.1. Subsystem Overview

The Transaction & Handover Subsystem is the core accountability engine of the UTM Borrow platform. It manages the complete lifecycle of resource sharing transactions between borrowers and lenders, from the initial request submission through to the final return verification. This subsystem ensures that every item exchange is securely tracked, digitally verified, and permanently audited for community safety and trust maintenance.

5.1.1. Purpose

Unlike informal peer-to-peer sharing via WhatsApp or Telegram, the Transaction & Handover Subsystem introduces structured, verifiable accountability mechanisms. It eliminates ambiguity surrounding item condition, return deadlines, and physical handover through cryptographic QR code scanning, creating an undeniable proof-of-transfer record that protects both lenders and borrowers.

5.1.2. Problem Statement

Campus resource sharing currently lacks formal accountability mechanisms, leading to disputes over damaged items, forgotten returns, and unresolved conflicts between lenders and borrowers. Without a centralized transaction management system, students resort to unstructured chat-based agreements that leave no audit trail. The absence of a formal handover verification protocol exposes both parties to scams and false damage claims. Additionally, community moderation is entirely absent, allowing inappropriate or prohibited items to circulate unchecked.

5.1.3. Subsystem Objectives

Digital Request Workflow: Enable structured, formally documented borrowing requests with automatic date validation and item status locking.

QR-Based Handover Verification: Implement cryptographic QR code generation and scanning to create undeniable proof-of-transfer at both pickup and return.

Transaction State Machine: Design a robust state machine managing transaction lifecycle (Pending → Approved → Borrowed → Completed).

Community Moderation: Provide students and administrators with tools to report, review, and enforce against inappropriate items and unsafe users.

Audit Logging: Maintain permanent, timestamped records of every transaction state change for compliance and dispute resolution.

5.1.4. SUBSYSTEM ARCHITECTURE OVERVIEW

5.1.4.1. Modules Structure

The Transaction & Handover Subsystem is organized into three integrated modules, each responsible for a distinct phase of the transaction lifecycle:

Module	Purpose & Scope
Request & Approval Workflow	Manages formal digital requests from borrowers, lender approval/rejection decisions, and optional request cancellation by either party before physical handover.
QR Verification Module	Generates unique cryptographic QR codes for secure handover and return verification, with timestamp logging for accountability.
Community Moderation & Reporting	Enables student reporting of inappropriate items, provides administrative review tools, and supports enforcement actions (item removal, user suspension).

5.1.4.2. Use Case Diagram

The following diagram illustrates the actors and their interactions across all three modules of the Transaction & Handover Subsystem.

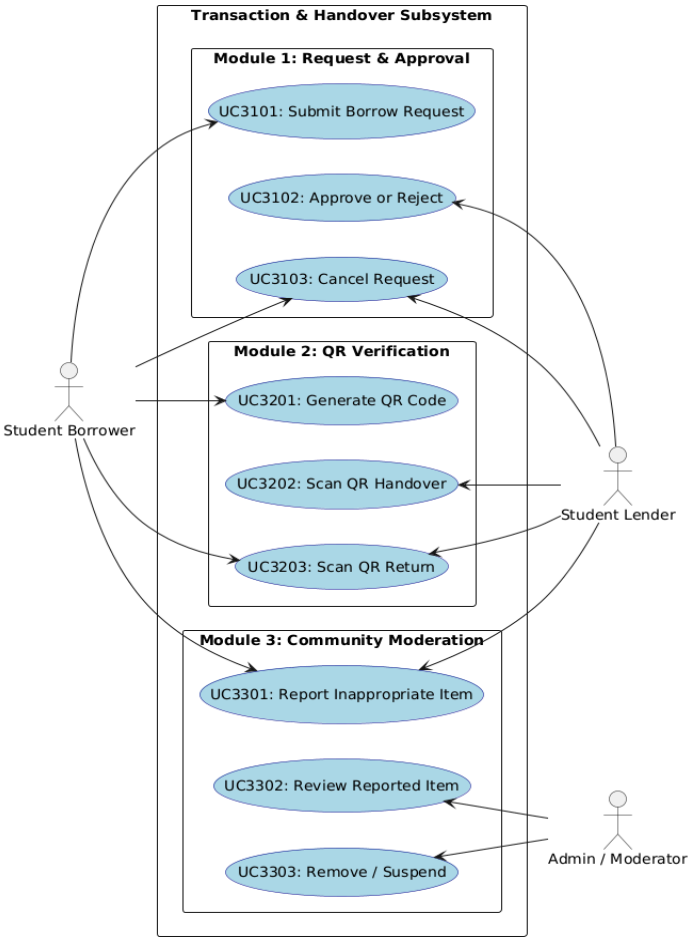


Figure 5.1.4: Use Case Diagram of Transaction & Handover Subsystem Architecture

5.2. Module 1: Request & Approval Workflow

5.2.1. Structure of Module 1:

The Request & Approval Workflow Module manages the initial digital agreement phase between borrowers and lenders. When a borrower discovers an item they wish to borrow, they formally submit a request specifying the desired borrow duration. The lender then reviews the borrower's profile and Trust Score before approving or rejecting the request. This module enforces date validation, automatic item status locking, and provides flexibility for either party to cancel before the physical meetup occurs.

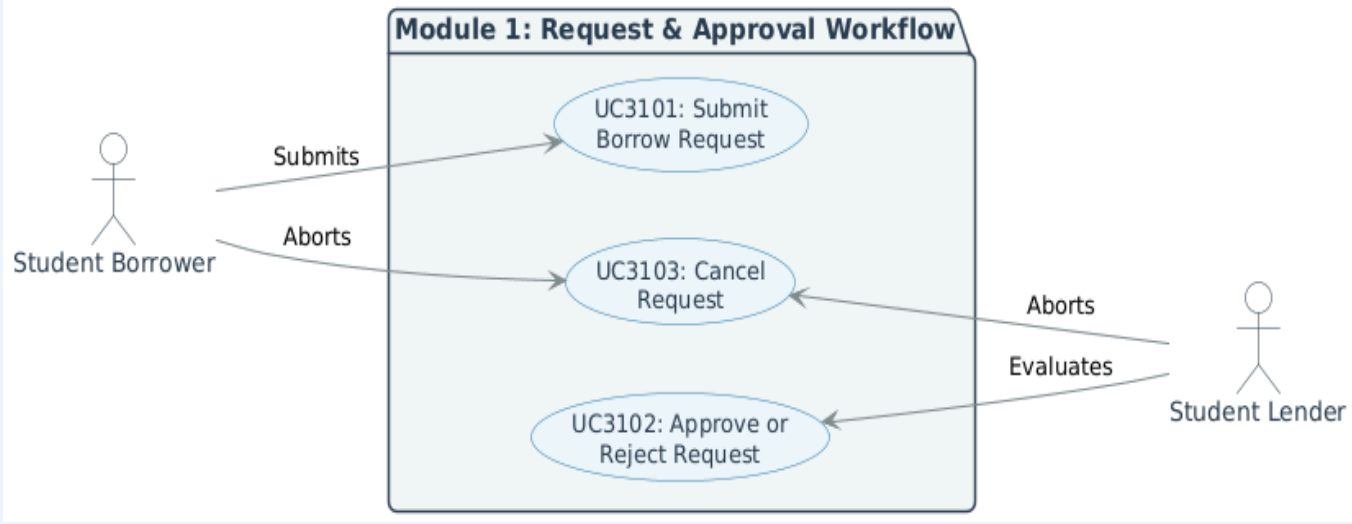


Figure 5.2: Package Diagram of Request & Approval Workflow Module

5.2.2. Specific Requirements

This module must maintain strict transaction state integrity. All date inputs must be validated to prevent past-dated or invalid duration requests. Upon successful request submission, the item's status must immediately transition from "Available" to "Pending" to prevent overlapping requests. The lender must receive real-time notifications of incoming requests. All state transitions (Pending → Approved or Rejected) must be logged with precise timestamps. Cancellation must be allowed only before QR scanning occurs, ensuring that physical handovers cannot be retroactively undone.

5.2.3. UC3101: Submit Borrow Request

5.2.3.1. Use Case Specifications

Use Case: Submit Borrow Request	ID: UC3101
Actors	Student Borrower (Borrower), System
Module	Request & Approval Workflow

Preconditions

1. The borrower must be logged into their verified UTM Borrow account.
2. The borrower is viewing an item whose current status is "Available".

Flow of Events

1. The borrower is on the Item Details screen viewing an item they want to borrow.
2. The borrower taps the "Request to Borrow" button at the bottom of the screen.
3. The system displays a request form prompting the borrower to select the required borrow duration (Start Date and End Date / Return Date).
4. The borrower inputs the required dates and taps "Send Request".
5. The system validates the selected dates.
6. The system records the request in the database and sets the transaction status to "Pending".
7. The system updates the item's status in the database from "Available" to "Pending" to temporarily lock it from other requests.
8. The system displays a success message: "Borrow Request sent successfully. Waiting for Lender's approval."
9. The system automatically navigates the borrower back to the Resource Catalog.
10. The system triggers a push notification to the item's owner (Lender) regarding the new pending request.

Alternate Flows

A1:

1. At step 5, the system detects the Return Date is before the Start Date, or the dates selected are in the past.
2. The system displays a validation error message: "Please select a valid borrowing duration."
3. The borrower must correct the dates before the request can be submitted.

A2:

1. At step 5, the system detects that the item's status has just changed to "Pending" or "Borrowed" (another user submitted a request moments earlier).
2. The system blocks the request creation and displays an error alert: "Sorry, this item was just requested by another student."
3. The borrower is returned to the Item Details screen showing the updated "Pending" status.

Post-conditions

1. A new transaction record is successfully created in the database with a "Pending" status.
2. The specific item is temporarily locked (status = "Pending") and cannot be requested by other users.
3. The Lender receives a push notification alerting them of a pending borrow request to review.

Table 5.2.3: Use Case Description of UC3101 — Submit Borrow Request

5.2.3.2. Sequence Diagram

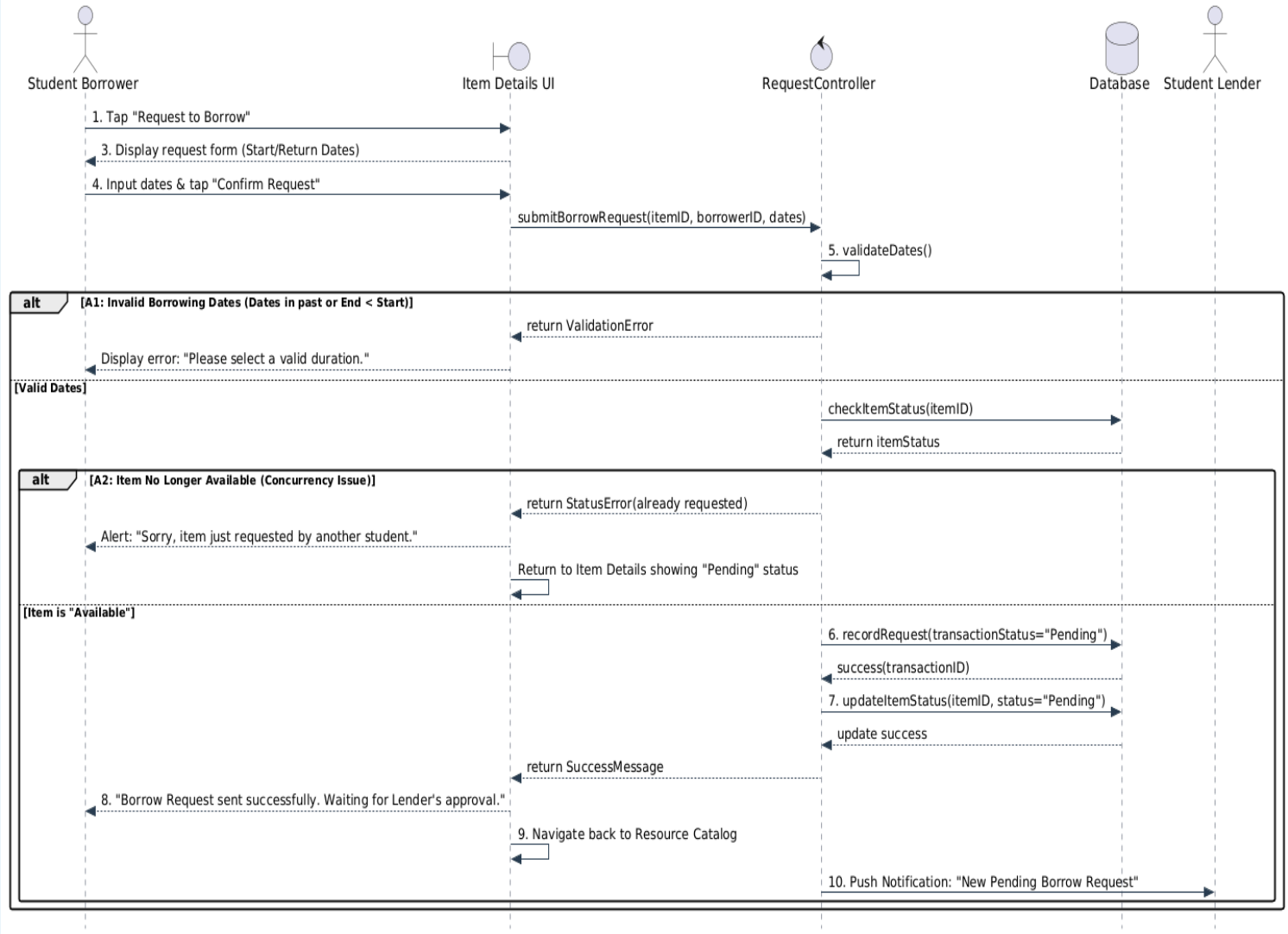


Figure 5.2.3: Sequence Diagram of UC3101 — Submit Borrow Request

5.2.4. UC3102: Approve or Reject Request

5.2.4.1. Use Case Specifications

Use Case: Approve or Reject Request	ID: UC3102
Actors	Student Lender (Lender), System
Module	Request & Approval Workflow

Preconditions

1. The Lender must be logged into their verified UTM Borrow account.
2. There is at least one active transaction record with a "Pending" status assigned to one of the Lender's listed items.

Flow of Events

1. The Lender navigates to the "User Activity Dashboard" and taps on the "Pending Requests" tab.
2. The system retrieves and displays all pending request details: Borrower's name, Borrower's Trust Score, requested item name, and specified borrow dates.
3. The Lender reviews the Borrower's profile information and taps the "Approve" button.
4. The system prompts a confirmation dialog: "Are you sure you want to approve this request?"
5. The Lender confirms the approval.
6. The system updates the transaction status in the database to "Approved".
7. The system maintains the item's status as "Pending" (physical handover has not yet occurred).
8. The system displays a success message: "Request Approved! You can now arrange the meetup with the borrower."
9. The system generates and sends an "Approved" push notification to the Borrower.

Alternate Flows

A1 — Lender Rejects:

1. At step 3, the Lender chooses to tap the "Reject" button instead of "Approve".
2. The system prompts a confirmation dialog: "Are you sure you want to reject this request?"
3. The Lender confirms the rejection.
4. The system updates the transaction status to "Rejected" in the database.
5. The system immediately reverts the item's status back to "Available" so other students can request it.
6. The system displays a success message: "Request Rejected."
7. The system sends a "Rejected" push notification to the Borrower.

A2 — Request Cancelled by Borrower:

1. At step 5, before the Lender finalizes the approval, the system detects that the Borrower has just cancelled the request.
2. The system blocks the approval action and displays: "This request is no longer valid — the borrower has cancelled it."
3. The system returns the Lender to the Dashboard.

Post-conditions

1. The transaction status is updated to either "Approved" or "Rejected".
2. If approved, both parties are notified and the process advances to the QR Verification phase.
3. If rejected, the item is restored to the Resource Catalog with an "Available" status.

Table 5.2.4: Use Case Description of UC3102 — Approve or Reject Request

5.2.4.2. Sequence Diagram

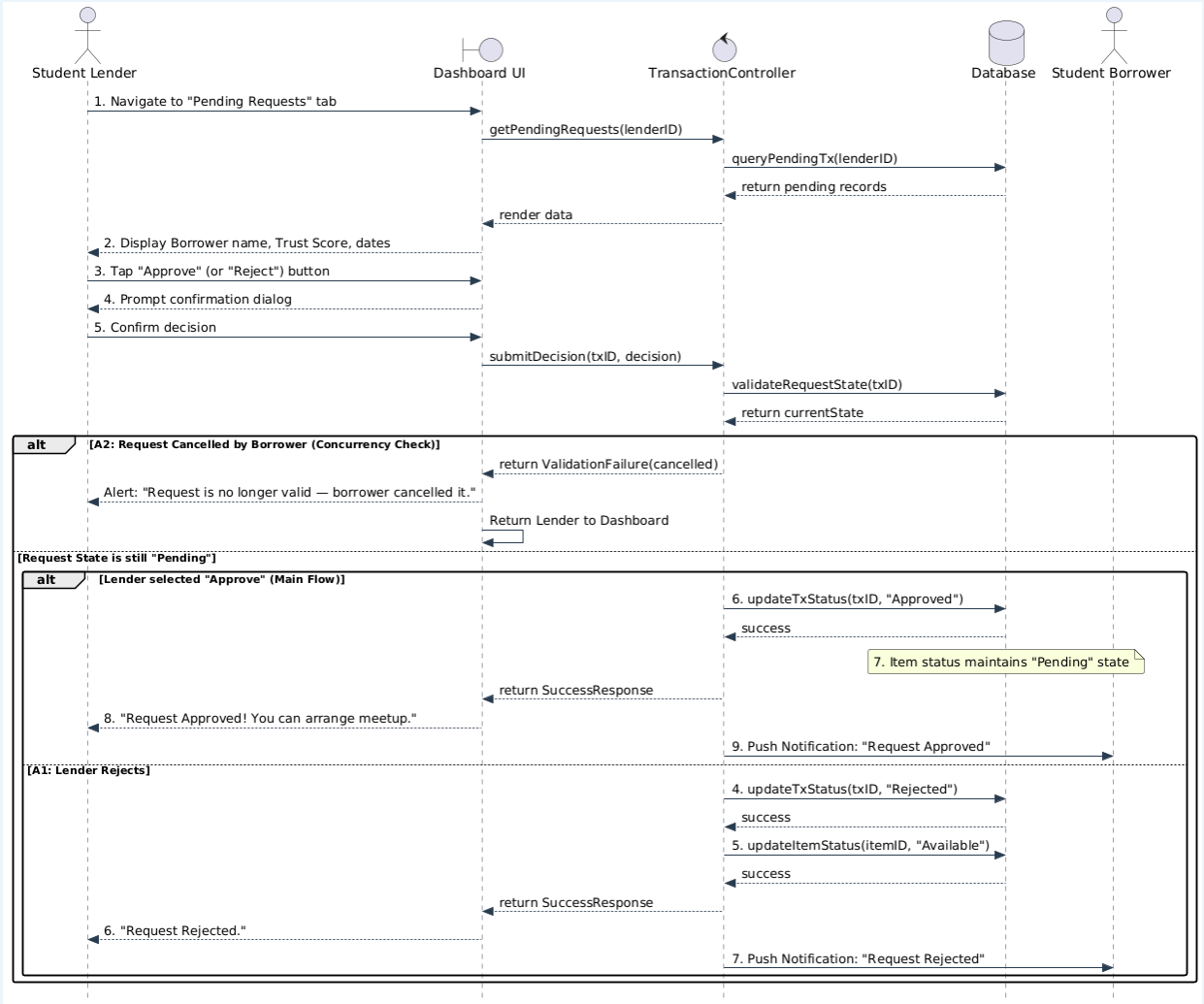


Figure 5.2.5: Sequence Diagram of UC3102 — Approve or Reject Request

5.2.5. UC3103: Cancel Request

5.2.5.1. Use Case Specifications

Use Case: Cancel Request	ID: UC3103
Actors	Student Borrower OR Student Lender, System
Module	Request & Approval Workflow

Preconditions

1. The user (Borrower or Lender) must be logged into their verified UTM Borrow account.
2. The transaction status is either "Pending" or "Approved", but the physical handover (QR scan) has NOT yet occurred.

Flow of Events

1. The user navigates to their "User Activity Dashboard".
2. The user taps on an active transaction they wish to cancel.
3. The system displays the full transaction details screen.
4. The user taps the "Cancel Request" button.
5. The system prompts a confirmation dialog: "Are you sure you want to cancel this transaction? This action cannot be undone."
6. The user confirms the cancellation.
7. The system updates the transaction status in the database to "Cancelled".
8. The system updates the associated item's status back to "Available".
9. The system displays a success message: "Transaction cancelled successfully."
10. The system automatically sends a push notification to the other party informing them of the cancellation.

Alternate Flows

A1 — Handover Already Completed:

1. At step 4, the system detects that the item's status has just changed to "Borrowed" (the QR code was scanned, physical handover already occurred).
2. The system immediately blocks the cancellation and hides the "Cancel Request" button.
3. The system displays an error alert: "You cannot cancel this transaction — the item has already been physically handed over."

A2 — User Cancels the Cancellation:

1. At step 6, the user taps "No" or dismisses the confirmation dialog.
2. The system closes the dialog without making any changes.
3. The transaction remains in its current active state.

Post-conditions

1. The transaction record is marked as "Cancelled" and archived in the transaction history.
2. The specific item is released back into the catalog with an "Available" status.
3. The other party receives a notification informing them of the cancellation.

Table 5.2.5: Use Case Description of UC3103 — Cancel Request

5.2.5.2. Sequence Diagram

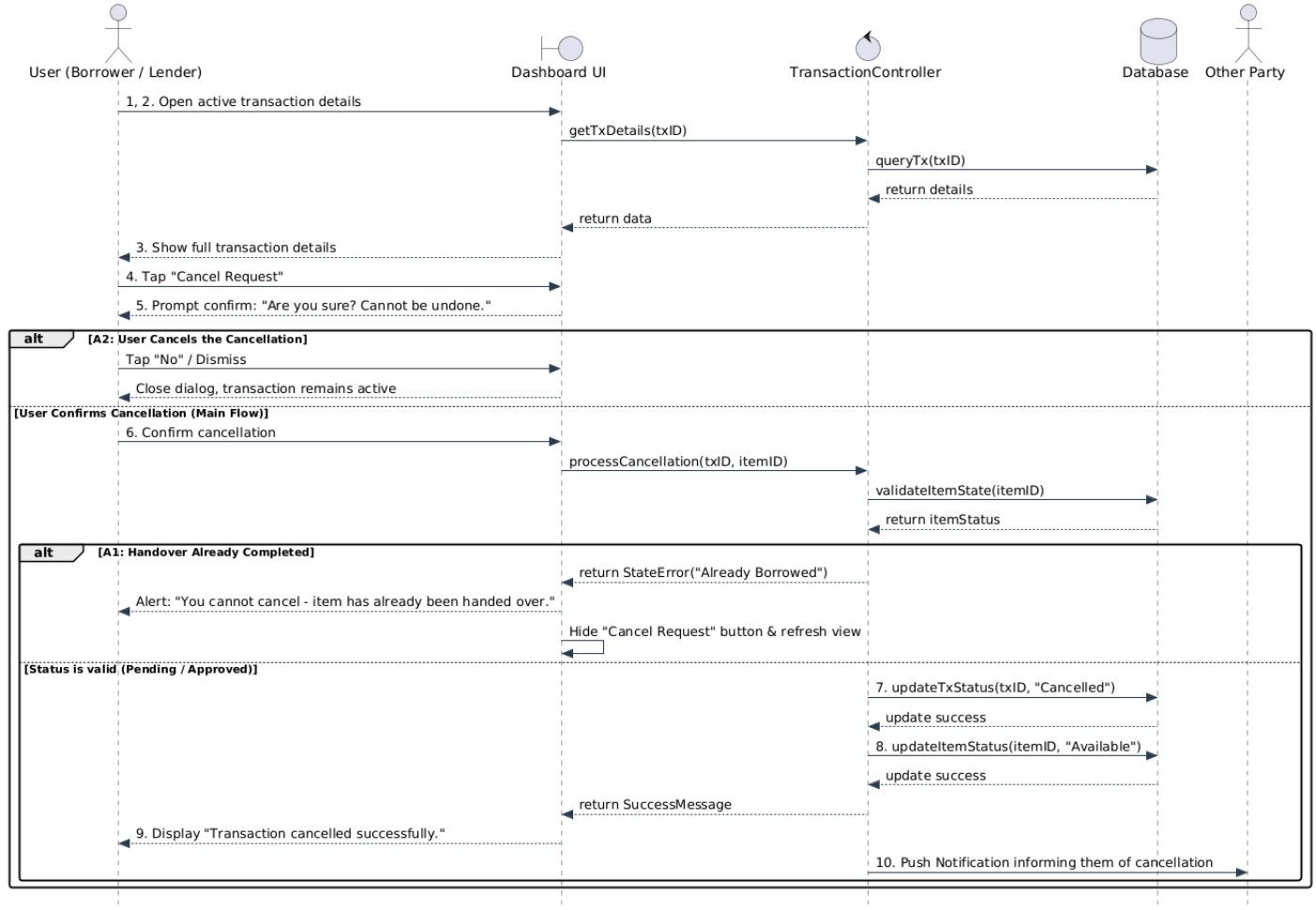


Figure 5.2.5: Sequence Diagram of UC3103 — Cancel Request

5.3. Module 2: QR Verification

5.3.1. Structure of Module 2:

The QR Verification Module is the accountability backbone of UTM Borrow. It generates unique, time-sensitive cryptographic QR codes linked to each approved transaction. During the physical meetup, the lender scans the borrower's QR code to record the exact handover moment, officially transitioning the item to "Borrowed" status. At the end of the borrow period, the same QR code is scanned again upon return, officially closing the transaction and returning the item to "Available" status. This dual-verification approach creates an immutable audit trail that eliminates disputes over item condition and return status.

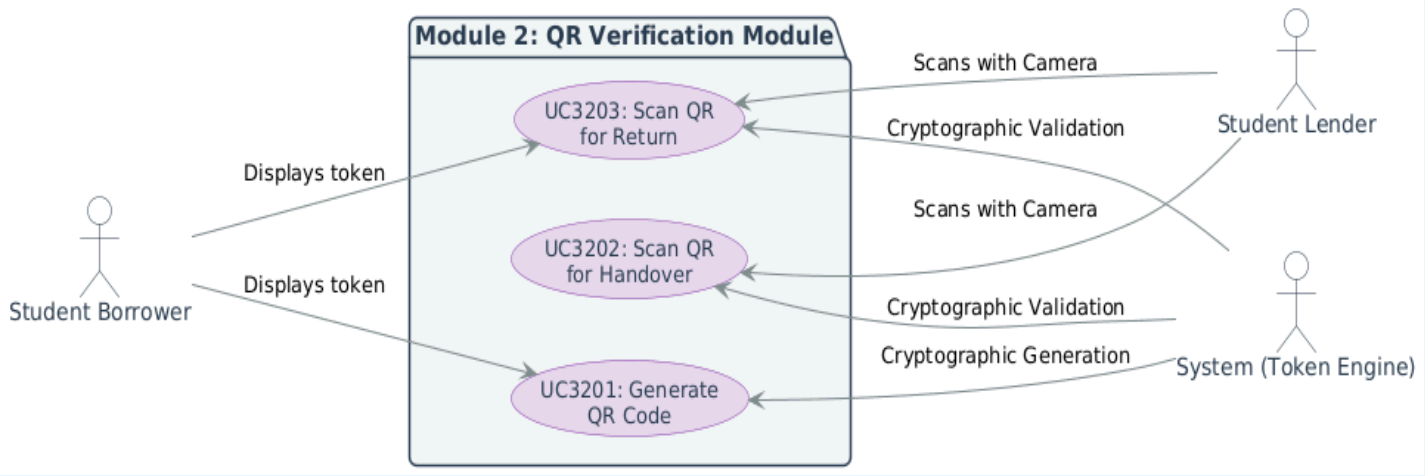


Figure 5.3: Package Diagram of QR Verification Module.

5.3.2. Specific Requirements

All QR codes must be cryptographically signed with transaction ID and borrower user ID to prevent spoofing. Token generation must be time-sensitive to prevent replay attacks. The system must maintain detailed timestamp logs for every handover and return event. Camera permissions must be gracefully handled with user-friendly error messages. Network failures during QR generation or validation must be recoverable with automatic retry mechanisms. Invalid or mismatched QR codes must be rejected with clear error messaging that guides users to rescan the correct code.

5.3.3. UC3201: Generate QR Code

5.3.3.1. Use Case Specifications

Use Case: Generate QR Code ID: UC3201	
Actors	Student Borrower (Borrower), System
Module	QR Verification
Preconditions	
1. The transaction status is "Approved" — the Lender has accepted the borrow request. 2. The Borrower and Lender have mutually agreed to a physical meeting time and location.	
Flow of Events	
1. The Borrower navigates to the "User Activity Dashboard" and opens the specific "Approved" transaction. 2. The system fetches the transaction data from the database. 3. The Borrower taps the "Show QR Code" button. 4. The system securely generates a unique, time-sensitive cryptographic QR code linked directly to the transaction ID and the Borrower's verified user ID. 5. The system displays the generated QR code on the Borrower's screen, accompanied by instructions: "Show this QR code to the Lender to confirm the handover."	
Alternate Flows	
A1 — Network Error During Generation: 1. The system fails to connect to the backend to securely sign and generate the token. 2. The system displays an error: "Network Error: Could not generate QR code. Please try again." 3. The system displays a "Retry" button for the Borrower to attempt regeneration.	
Post-conditions	
1. A secure, unique cryptographic QR code is displayed on the Borrower's device screen, ready to be scanned by the Lender.	

Table 5.3.4: Use Case Description of UC3201 — Generate QR Code

5.3.3.2. Sequence Diagram

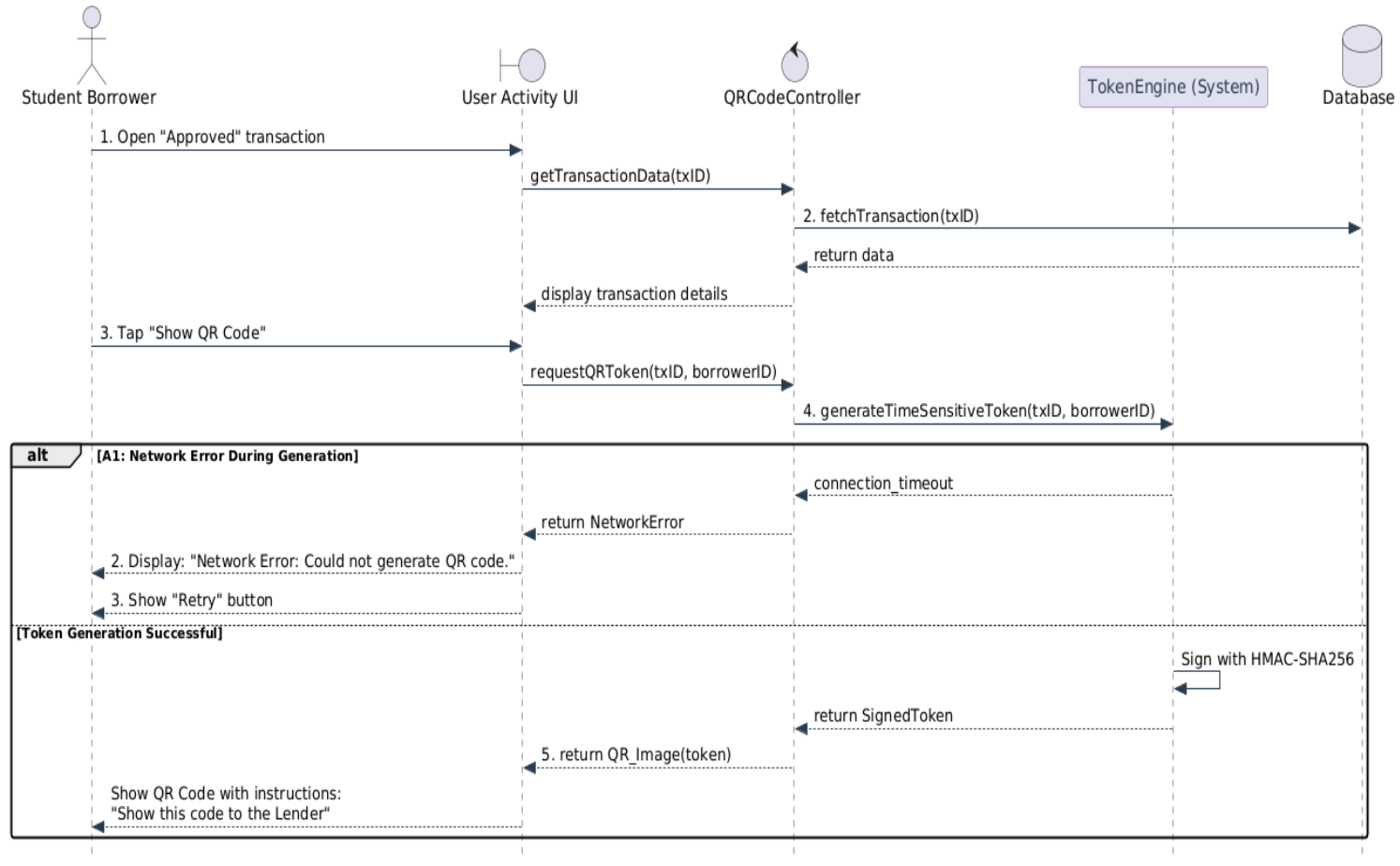


Figure 5.3.4: Sequence Diagram of UC3201 — Generate QR Code

5.3.4. UC3202: Scan QR for Handover (Borrowing item)

5.3.4.1. Use Case Specifications

Use Case: Scan QR for Handover	ID: UC3202
Actors	Student Lender (Lender), System
Module	QR Verification

Preconditions

1. The transaction status is "Approved".
2. The Lender has granted the application camera access on their device.
3. The Borrower is physically present and displaying the QR code generated in UC3201.

Flow of Events

1. The Lender navigates to the active transaction in their "User Activity Dashboard" and taps "Scan Handover QR".
2. The system activates the device camera for QR scanning.
3. The Lender points their camera at the QR code displayed on the Borrower's device.
4. The system captures the QR token and securely validates it against the database.
5. The token MATCHES the expected transaction ID and is confirmed valid.
6. The system updates the transaction status in the database from "Approved" to "Borrowed".
7. The system records the exact handover timestamp.
8. The system displays a success alert to the Lender: "Handover Verified! Item is now marked as Borrowed."
9. The system pushes a real-time notification to the Borrower confirming the item is legally in their possession.

Alternate Flows

A1 — Invalid or Mismatched QR Code:

1. At step 5, the token extracted does NOT match the expected transaction, or the QR belongs to a different item.
2. The system displays an error: "Invalid QR Code. Please ensure the Borrower is showing the correct transaction QR."
3. The Lender is prompted to rescan the correct QR code.

A2 — Camera Permission Denied:

1. At step 2, the Lender has denied camera permissions on their device.
2. The system displays an alert: "Camera permission is required to scan QR codes. Please enable it in your device settings."

Post-conditions

1. The item's status in the database changes from "Pending" to "Borrowed".
2. A digital lease cycle officially begins, securing accountability for both parties.
3. A handover timestamp is permanently logged for audit purposes.

Table 5.3.5: Use Case Description of UC3202 — Scan QR for Handover

5.3.4.2. Sequence Diagram

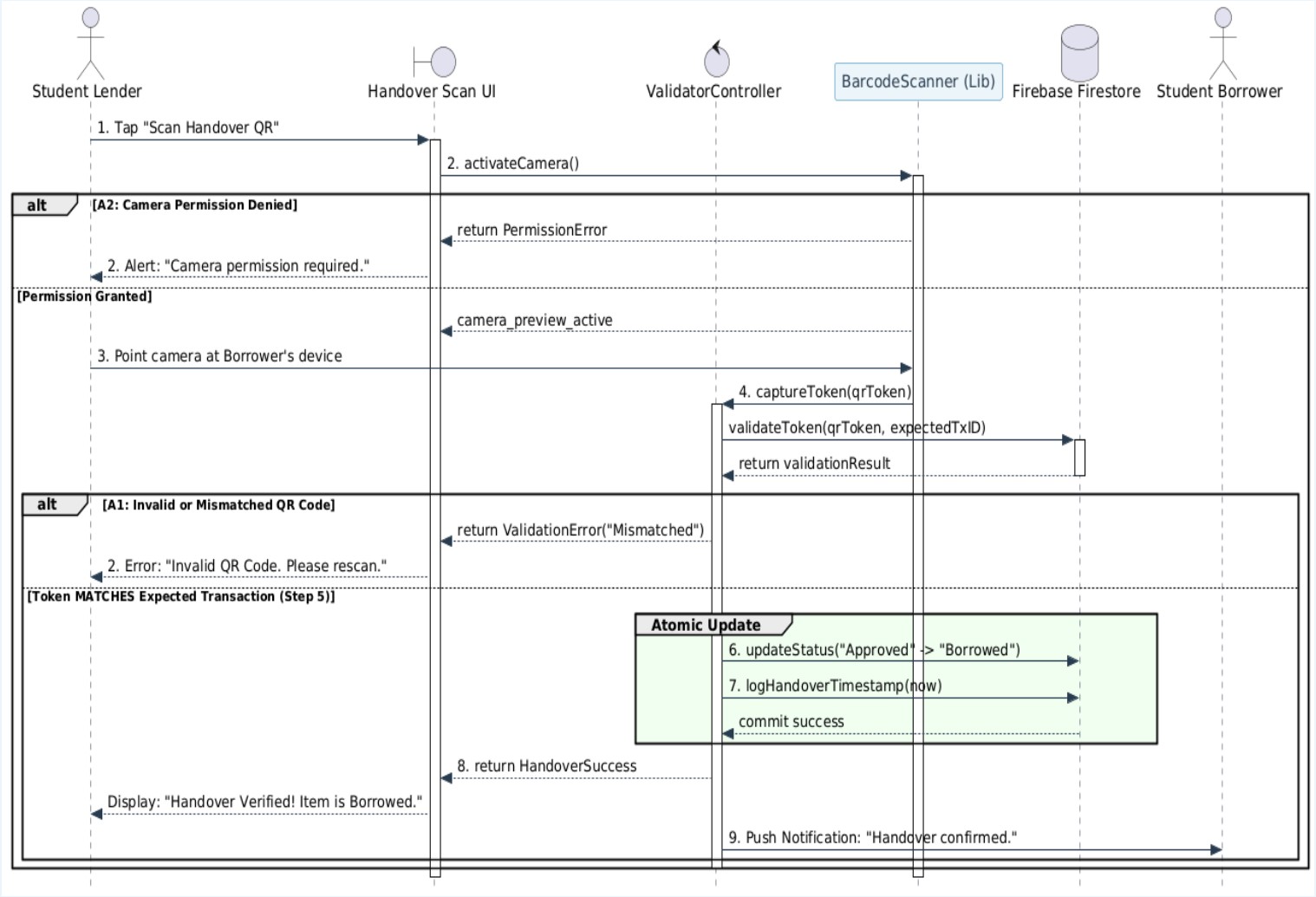


Figure 5.3.5: Sequence Diagram of UC3202 — Scan QR for Handover

5.3.5. UC3203: Scan QR for Return (Returning item)

5.3.5.1. Use Case Specifications

Use Case: Scan QR for Return	ID: UC3203
Actors	Student Lender (Lender), Student Borrower (Borrower), System
Module	QR Verification

Preconditions

1. The transaction status is "Borrowed" and the Borrower is ready to physically return the item.
2. The Lender has granted the application camera access on their device.
3. The Borrower is physically present and can display the original transaction QR code.

Flow of Events

1. The Borrower navigates to the active transaction in their "User Activity Dashboard" and taps "Show Return QR".
2. The system re-displays the original cryptographic QR code for that transaction.
3. The Lender navigates to the same active transaction in their dashboard and taps "Scan Return QR".
4. The system activates the Lender's device camera.
5. The Lender scans the QR code displayed on the Borrower's device.
6. The system captures and validates the QR token against the database, confirming it matches the transaction ID and has a "Borrowed" status.
7. The token is confirmed VALID and the statuses match.
8. The system updates the transaction status in the database from "Borrowed" to "Completed".
9. The system updates the item's status back to "Available", making it visible in the Resource Catalog for future borrowers.
10. The system records the exact return timestamp.
11. The system displays a success alert: "Return Verified! Item is now marked as Available."
12. The system immediately prompts both the Lender and Borrower to submit a Trust Score rating for each other.
13. Both users receive a notification confirming the successful completion of the transaction.

Alternate Flows

A1 — Invalid or Mismatched QR Code:

1. At step 6, the extracted token does NOT match the active transaction ID.
2. The system displays an error: "Invalid QR Code. Please ensure the correct transaction QR is shown."
3. The Lender is prompted to retry the scan.

A2 — Camera Permission Denied:

1. At step 4, the Lender's camera permission is denied.
2. The system displays: "Camera permission is required. Please enable it in your device settings."

A3 — Transaction Already Completed:

1. At step 6, the system detects the transaction status is already "Completed".
2. The system blocks the scan and displays: "This transaction has already been closed."

Post-conditions

1. The transaction status is updated to "Completed" and archived in both users' Transaction History.
2. The item's status is restored to "Available" and re-appears in the Resource Catalog.
3. The exact return timestamp is recorded for auditing purposes.

4. Both users are prompted to rate each other's Trust Score to maintain the community reputation system.

Table 5.3.6: Use Case Description of UC3203 — Scan QR for Return

5.3.5.2. Sequence Diagram

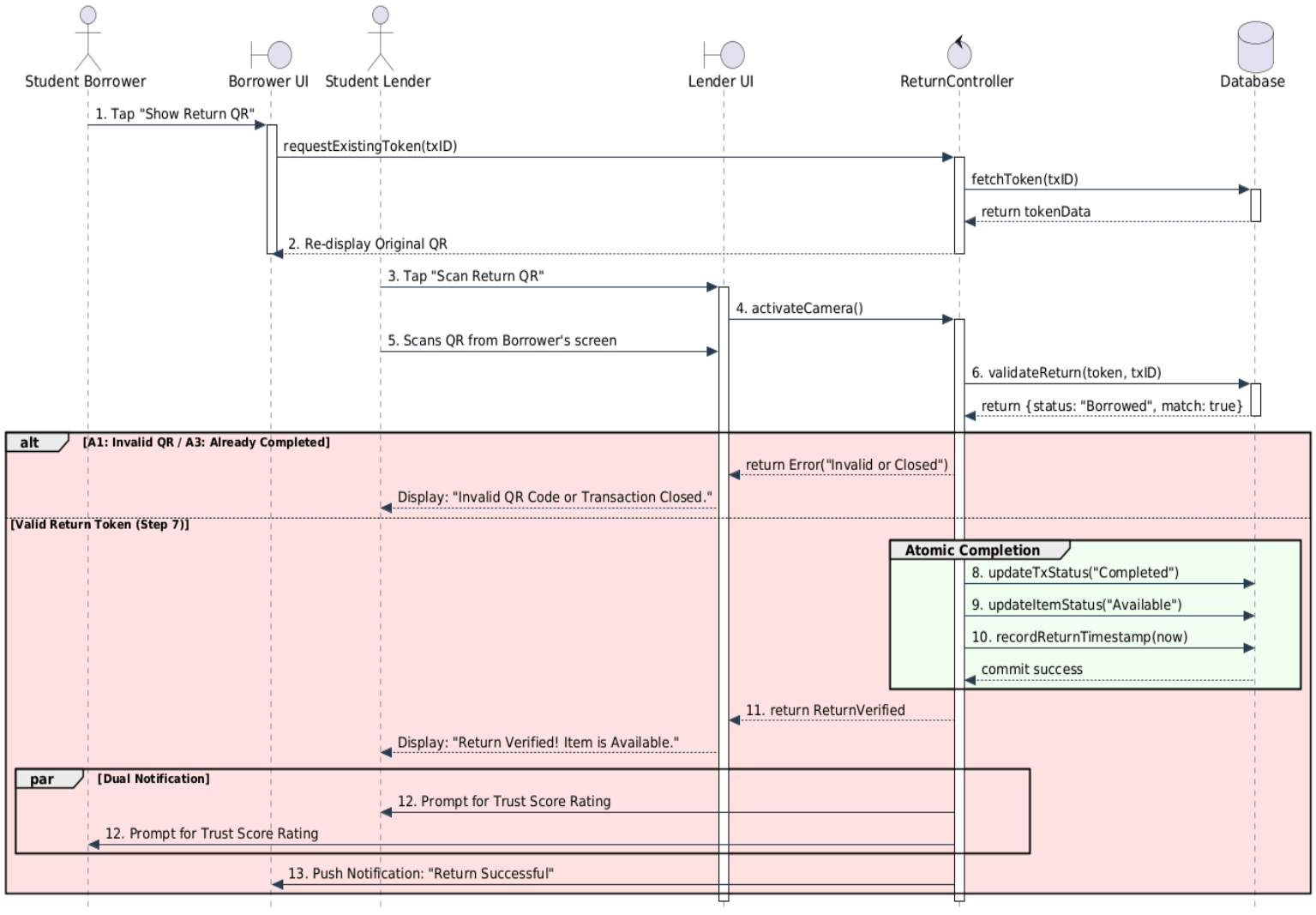


Figure 5.3.6: Sequence Diagram of UC3203 — Scan QR for Return

5.4. Module 3: Community Moderation & Reporting

5.4.1.1. Structure of Module 3:

The Community Moderation & Reporting Module provides essential safeguards to maintain UTM Borrow as a trusted, safe resource-sharing environment. It empowers verified students to report listings that violate platform guidelines—such as prohibited substances, scam listings, severely damaged items, or inappropriate content. Administrators access a centralized moderation dashboard to review reported items, examine lender profiles, assess transaction histories, and make evidence-based enforcement decisions. The module supports two enforcement actions: permanent item removal or temporary user account suspension.

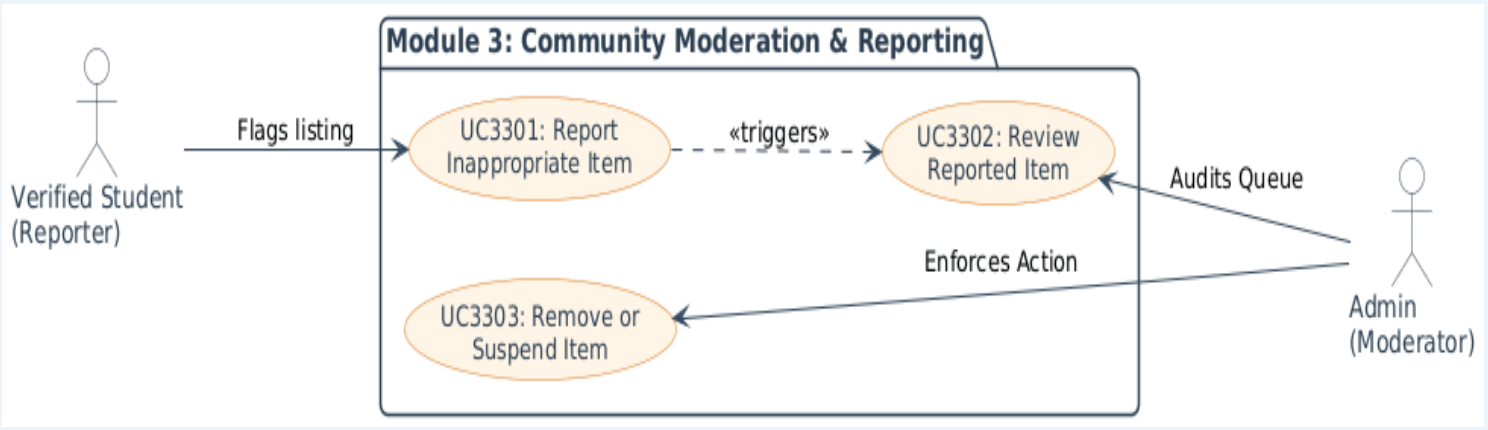


Figure 5.4: Package Diagram of Community Moderation & Reporting Module.

5.4.1.2. Specific Requirements

Report submission must validate that a category is selected before acceptance. Duplicate reports from the same user for the same item must be prevented with clear messaging. The moderation dashboard must display comprehensive report context: item metadata, lender profile and Trust Score, reporter's claim with optional description, and transaction history. Admins must be able to dismiss reports with optional notes, or escalate to enforcement action. All moderation decisions must be logged with admin ID, decision type, reason, and timestamp for audit compliance. Suspended users must have their active listings hidden from the catalog during the suspension period.

5.4.2. UC3301: Report Inappropriate Item

5.4.2.1. Use Case Specifications

Use Case: Report Inappropriate Item	ID: UC3301
Actors	Verified Student (Reporter), System
Module	Community Moderation & Reporting

Preconditions

1. The Reporter must be logged into their verified UTM Borrow account.
2. The Reporter is viewing a specific item listing in the Resource Catalog or is involved in an active transaction.
3. The Reporter has not already submitted a report for this specific item.

Flow of Events

1. The Reporter navigates to the item listing they wish to report.
2. The Reporter taps the "Report" icon (flag icon) on the item listing page.
3. The system displays a Report Form with the following category options: (1) Prohibited or Illegal Item, (2) False Information / Scam Listing, (3) Severely Damaged or Dangerous Item, (4) Inappropriate or Offensive Content, (5) Item Not Available — False Listing, (6) Other.
4. The Reporter selects the most appropriate report category.
5. The Reporter optionally enters additional details in the provided description text field.
6. The Reporter taps "Submit Report".
7. The system validates that a report category has been selected.
8. The system records the report in the moderation database, linking it to the item ID, reporter's user ID, selected category, optional description, and submission timestamp.
9. The system internally flags the item for administrative review.
10. The system displays a confirmation message: "Report submitted. Our moderation team will review it shortly. Thank you for keeping UTM Borrow safe."

Alternate Flows

A1 — No Category Selected:

1. At step 7, the Reporter has not selected any report category.
2. The system highlights the category selection field in red and displays: "Please select a report category to proceed."
3. The Reporter must select a category before the form can be submitted.

A2 — Item Already Reported by This User:

1. At step 2, the system detects that this Reporter has already submitted a report for this specific item.
2. The system displays: "You have already submitted a report for this item. Our team is currently reviewing it."
3. The Reporter cannot submit a duplicate report for the same item.

A3 — Network Error During Submission:

1. At step 8, a network connection failure occurs while the system attempts to save the report.
2. The system displays: "Failed to submit report. Please check your connection and try again."
3. The Reporter can retry once connectivity is restored.

Post-conditions

1. A report record is successfully created and stored in the moderation database.
2. The item is internally flagged and added to the administrator's moderation review queue.
3. The Reporter receives an on-screen confirmation of the successful submission.
4. The item continues to be visible to other users until an administrator takes action.

Table 5.4.2: Use Case Description of UC3301 — Report Inappropriate Item

5.4.2.2. Sequence Diagram

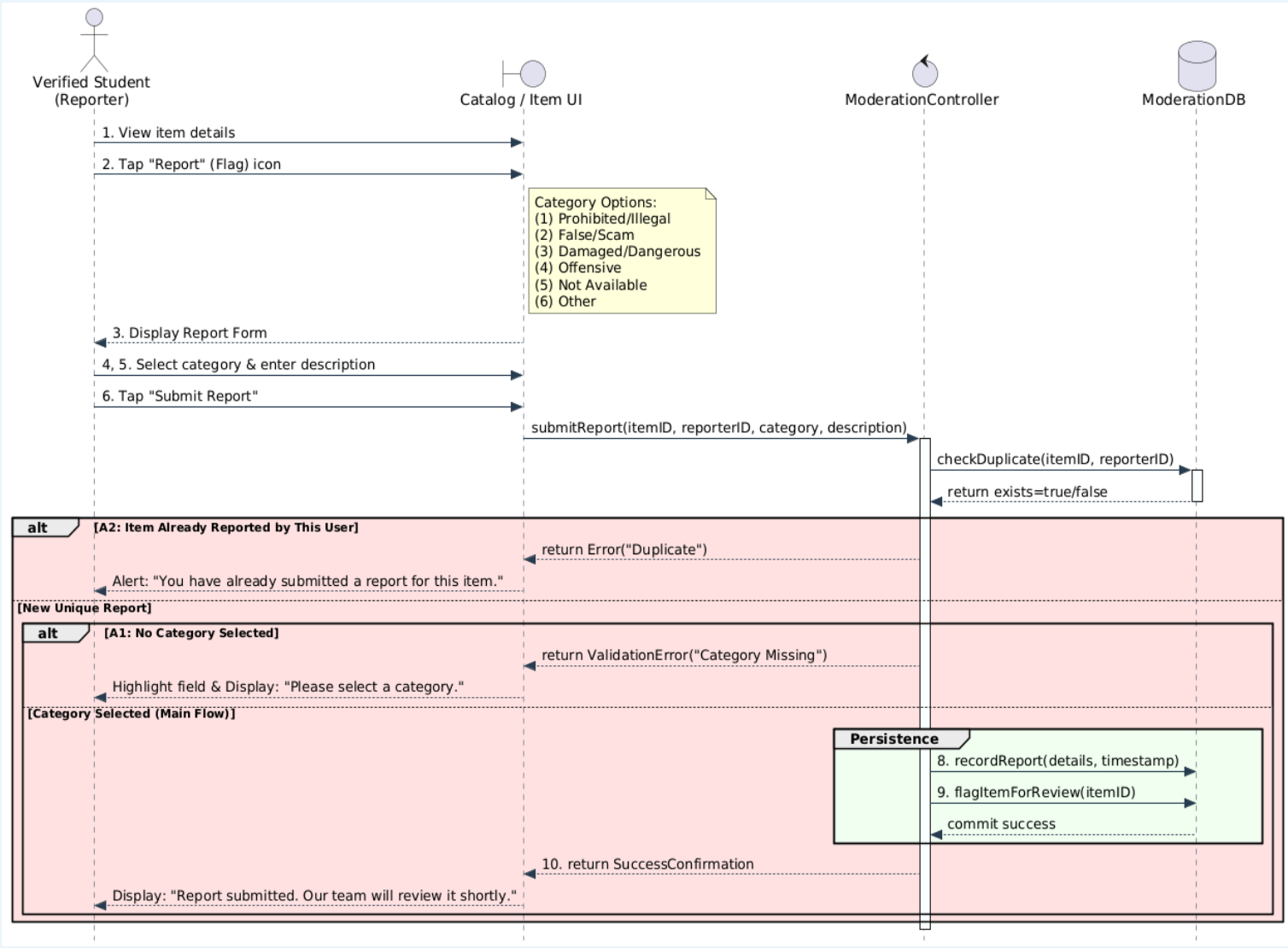


Figure 5.4.2: Sequence Diagram of UC3301 — Report Inappropriate Item

5.4.3. UC3302: Review Reported Item (Admin)

5.4.3.1. Use Case Specifications

Use Case: Review Reported Item	ID: UC3302
Actors	Admin / Moderator, System
Module	Community Moderation & Reporting

Preconditions

1. The Admin must be logged into the UTM Borrow system with administrative moderator privileges.
2. There is at least one flagged item pending review in the moderation queue.

Flow of Events

1. Admin navigates to the "Moderation Dashboard" from the Admin interface.
2. The system retrieves all pending reports from the database and displays them in a moderation queue, ordered by submission date (oldest first).
3. The system displays summary statistics: Total Pending Reports and count of High-Priority Flags.
4. Admin selects a report from the queue by tapping on it.
5. The system displays the full Report Detail View containing: reported item info (title, photos, category, condition), lender's profile and Trust Score, reporter's claim (selected category and optional description), item's current status and transaction history, and the report submission timestamp.
6. Admin thoroughly reviews all available information and evidence.
7. Admin decides to take enforcement action or dismiss the report as unfounded.
8. If Admin chooses to dismiss: Admin taps "Dismiss Report" and optionally enters a dismissal note.
9. The system prompts a confirmation dialog: "Are you sure you want to dismiss this report?"
10. Admin confirms the dismissal.
11. The system updates the report status to "Dismissed — No Action Required" and logs the decision.
12. The system sends a notification to the original Reporter informing them the report was reviewed.
13. The dismissed report is moved out of the active queue into the audit log.

Alternate Flows

A1 — No Reports in Queue:

1. At step 2, the moderation queue is empty.
2. The system displays: "No pending reports at this time. The platform is all clear!"

A2 — Report Already Actioned:

1. At step 4, the selected report has already been processed by another administrator.
2. The system displays: "This report has already been reviewed and actioned."
3. Admin is returned to the moderation queue.

A3 — Server Error Loading Details:

1. At step 5, the system fails to retrieve the full report details due to a server error.
2. The system displays: "Failed to load report details. Please try again."
3. Admin can tap Retry to re-attempt loading the details.

Post-conditions

1. The report status is updated (either "Dismissed" or actioned via UC3303).
2. The admin's decision, reviewer ID, and timestamp are logged for full audit trail compliance.

3. The original reporter is notified of the review outcome.
4. The moderation dashboard statistics are automatically updated to reflect the current queue.

Table 5.4.3: Use Case Description of UC3302 — Review Reported Item

5.4.3.2. Sequence Diagram

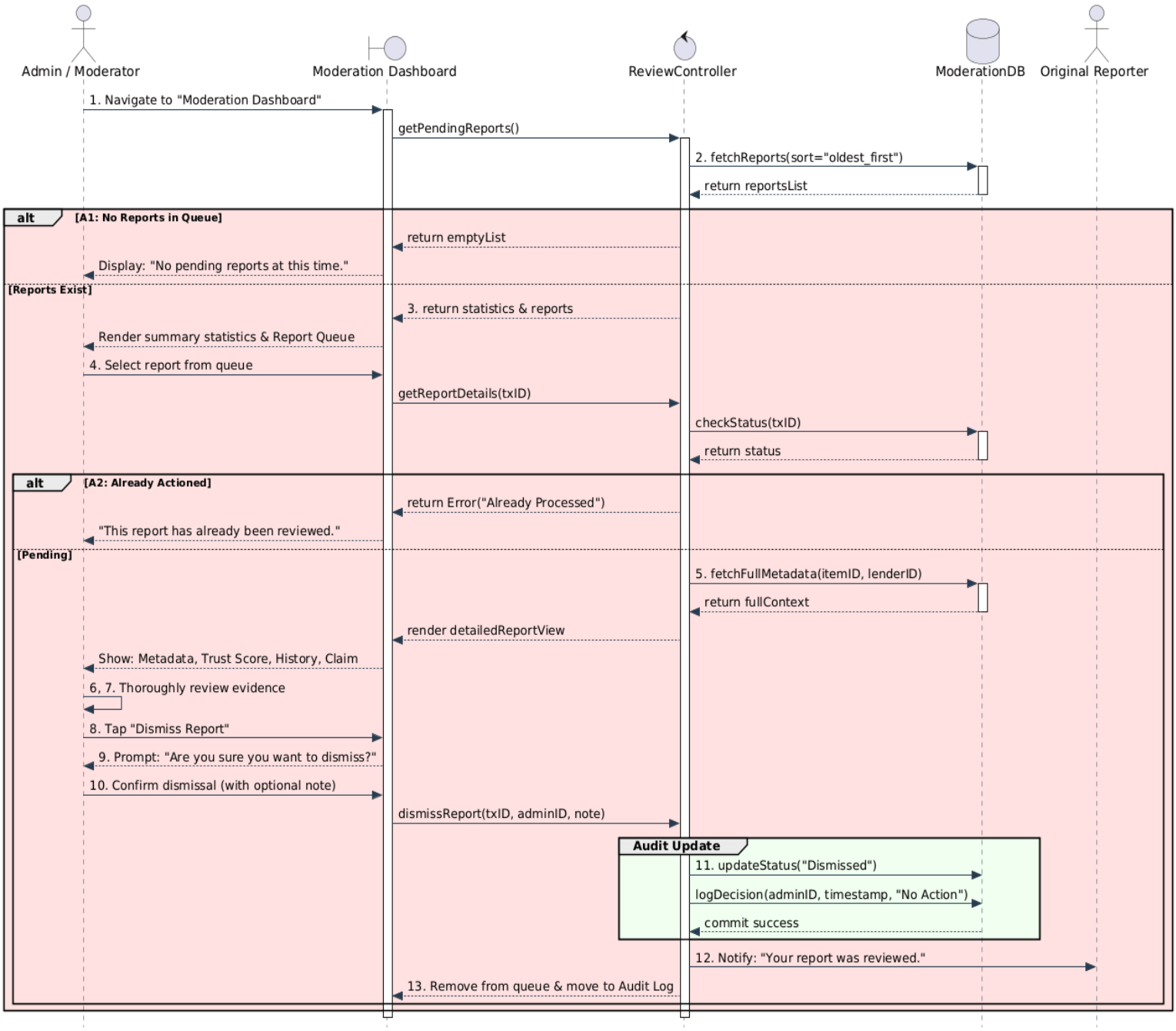


Figure 5.4.3: Sequence Diagram of UC3302 — Review Reported Item

5.4.4. UC3303: Remove/Suspend Item (Admin)

5.4.4.1. Use Case Specifications

Use Case: Remove or Suspend Item (Admin)	ID: UC3303
Actors	Admin / Moderator, System
Module	Community Moderation & Reporting

Preconditions

1. Admin is logged in with moderator / administrative privileges.
2. Admin has reviewed the flagged report via UC3302 and determined that enforcement action is necessary.
3. The reported item or user account still exists and is active in the system.

Flow of Events

1. Admin is viewing the Report Detail View (continued from UC3302, Step 6).
2. Admin taps the "Take Enforcement Action" button.
3. The system displays an Action Selection menu with two distinct options: (A) "Remove Item" — permanently removes the listing from the catalog, or (B) "Suspend User Account" — disables the user's ability to list or borrow for a specified period.
4. OPTION A — Remove Item: Admin selects "Remove Item". The system displays a confirmation dialog: "Permanently remove this item? This action cannot be undone." Admin confirms. The system sets the item status to "Removed" and immediately hides the listing from the Resource Catalog.
5. OPTION B — Suspend User Account: Admin selects "Suspend User Account". The system displays suspension duration options (3 Days, 7 Days, 30 Days, Permanent). Admin selects the duration and confirms. The system updates the user's account status to "Suspended" for the selected period.
6. The system logs the enforcement action with the admin's ID, action type, and exact timestamp.
7. The system notifies the affected user via push notification with an appropriate message.
8. The system updates the report status to "Actioned" and moves it to the audit log.
9. The system displays a success confirmation to the Admin: "Enforcement action completed successfully."

Alternate Flows

A1 — Item Already Removed:

1. At step 4 (Option A), the system detects the item has already been removed.
2. The system displays: "This item has already been removed from the catalog."

A2 — User Already Suspended:

1. At step 5 (Option B), the system detects the user account is already in a suspended state.
2. The system displays: "This user is already suspended. Would you like to modify the suspension period?"
3. Admin can choose to extend or shorten the existing suspension.

A3 — Admin Cancels the Action:

1. At any confirmation dialog, Admin taps "Cancel".
2. No changes are made to the item or user account.
3. Admin is returned to the Report Detail View.

A4 — Active Transaction Conflict:

1. At step 4 (Option A), the system detects the item has an active "Borrowed" status (it is currently on loan).

2. The system warns: "This item is currently on loan. Removing it will affect an active transaction. Confirm to proceed anyway?"
3. Admin can confirm to override or cancel to wait for the transaction to complete.

Post-conditions

1. If "Remove Item": The item is permanently hidden from the catalog and is no longer visible or accessible to any user.
2. If "Suspend Account": The user's account is suspended for the specified period; all their active listings are hidden and they cannot submit new requests.
3. The enforcement action is fully logged with admin ID, timestamp, and action type for compliance and audit purposes.
4. The affected user is notified of the specific enforcement action taken against their account or listing.
5. The report status is updated to "Actioned" and archived in the permanent audit log.

Table 5.4.4: Use Case Description of UC3303 — Remove or Suspend Item (Admin)

5.4.4.2. Sequence Diagram

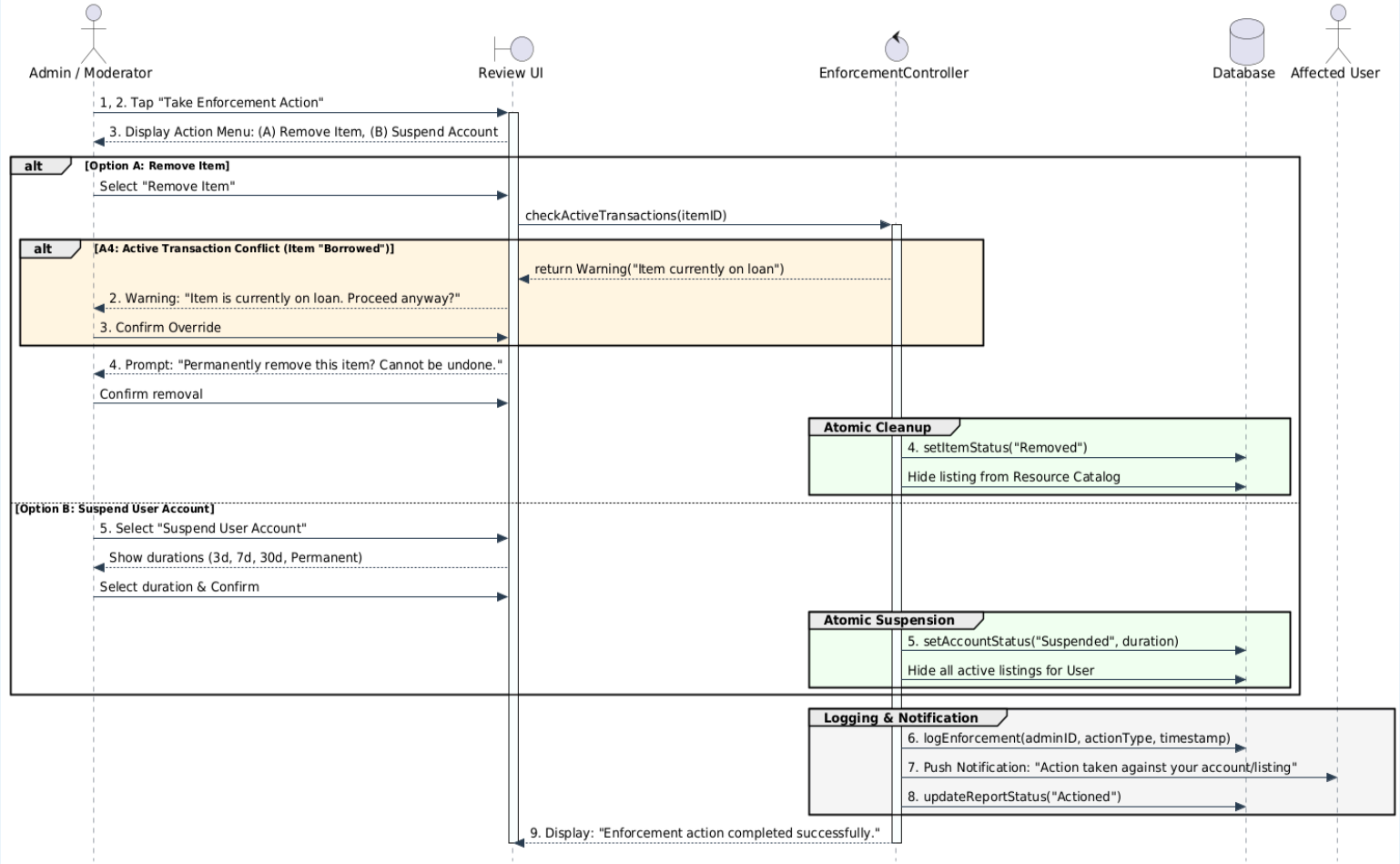


Figure 5.4.4: Sequence Diagram of UC3303 — Remove or Suspend Item

5.5. Technical Specifications

5.5.1. Non-Functional Requirements

5.5.1.1. Performance

1. QR code generation must complete within 2 seconds.
2. QR code scanning and validation must complete within 3 seconds.
3. Moderation dashboard must load report lists within 4 seconds, even with 500+ pending reports.
4. Push notifications must be delivered within 10 seconds of system events.

5.5.1.2. Security

1. All QR codes must be cryptographically signed using HMAC-SHA256 or equivalent.
2. Transaction IDs must be globally unique (UUID v4 format) to prevent ID guessing.
3. Token expiration must occur within 24 hours to prevent replay attacks
4. All moderation actions must be logged with role-based access control (RBAC).
5. User passwords must never be displayed or recoverable; only password resets are allowed.

5.5.1.3. Reliability & Data Integrity

1. Transaction state changes must be atomic; partial updates are not allowed.
2. All timestamps must use UTC timezone to prevent ambiguity across time zones.
3. Moderation decisions must be immutable; once logged, decisions cannot be altered (only new actions may be added).
4. Database backups must occur daily, with recovery time objective (RTO) of 4 hours.

5.5.1.4. Usability

1. QR scanning interface must work with device lighting conditions ranging from dim indoor to bright outdoor.
2. Error messages must be user-friendly, non-technical, and action-oriented (e.g., "Show the correct QR code" instead of "Token validation failed").
3. The platform must implement distinct visual color-coding for status badges (e.g., Yellow for Pending, Green for Borrowed, Red for Cancelled) to prevent user confusion."
4. Moderation dashboard must be accessible to admins with minimal training.

5.5.2. Sub-System Dependencies

1. Firebase Firestore for real-time transaction state synchronization across devices.
2. Firebase Push Notifications for sending alerts to lenders and borrowers.
3. Camera/QR scanning library compatible with both iOS and Android (e.g., BarcodeScanner).
4. Time-based cryptographic token generation using Node.js crypto library or equivalent.

5.6. Sub-System : CONCLUSION

The Transaction & Handover Subsystem represents the critical accountability layer of UTM Borrow. By implementing structured request workflows, cryptographic QR verification, and community-driven moderation, it transforms campus resource sharing from an informal, untracked activity into a secure, verifiable ecosystem. Every item exchange is now anchored in permanent audit logs, timestamp evidence, and role-based governance, ensuring that both lenders and borrowers can share with confidence and trust.

The three modules—Request & Approval Workflow, QR Verification, and Community Moderation & Reporting—work in concert to address the core problems identified in the project proposal: absence of accountability, lack of trust infrastructure, and absence of community safety oversight. Together, they establish UTM Borrow as a credible, trustworthy platform aligned with university values of transparency, sustainability, and student welfare.

5.6.1. Future Enhancements

1. Integration with digital damage claim submission forms during QR return scanning.
2. AI-powered photo analysis to automatically detect item condition changes.
3. Reputation decay mechanics to prevent lenders from inflating ratings through collusion.
4. Integration with campus mail systems for automated pickup/return reminders.
4. Escalation workflows for disputes that involve damage claims.